

★ WEEK 1 · PART 1 of 3

From Prices to Returns

The Language of Financial Markets

Simple vs log · arithmetic vs geometric · when to use which · what can go wrong

Before any model, we must speak the market's native unit: the return. This part builds that vocabulary carefully — every formula, its assumptions, and its traps.

1

PART 1 • RETURNS

Returns: The Building Block of Everything

Prices are what we observe. Returns are what we model. This part is the foundation the whole course stands on.

Why We Model Returns, Not Prices

Three reasons a professional never models the raw price series

1

Prices are non-stationary — returns are (nearly) stationary

Price levels trend and drift with no fixed mean or variance. Most statistical tools assume a stable distribution; returns provide it.

2

Returns are comparable across assets and scale

A \$10 move means one thing on a \$50 stock and another on a \$5,000 one. A 2% return means the same thing everywhere.

3

Returns compound cleanly — prices do not

Growth of wealth is a product of gross returns. Working in returns lets us aggregate over time and across a portfolio with simple algebra.

Rule of thumb: observe prices, but always convert to returns before you analyze.

What Is a Financial Time Series?

A single realized path drawn from an underlying random process

A time series is a sequence of random variables ordered in time. The data we see is ONE realized path — not the process itself.

The mental shift

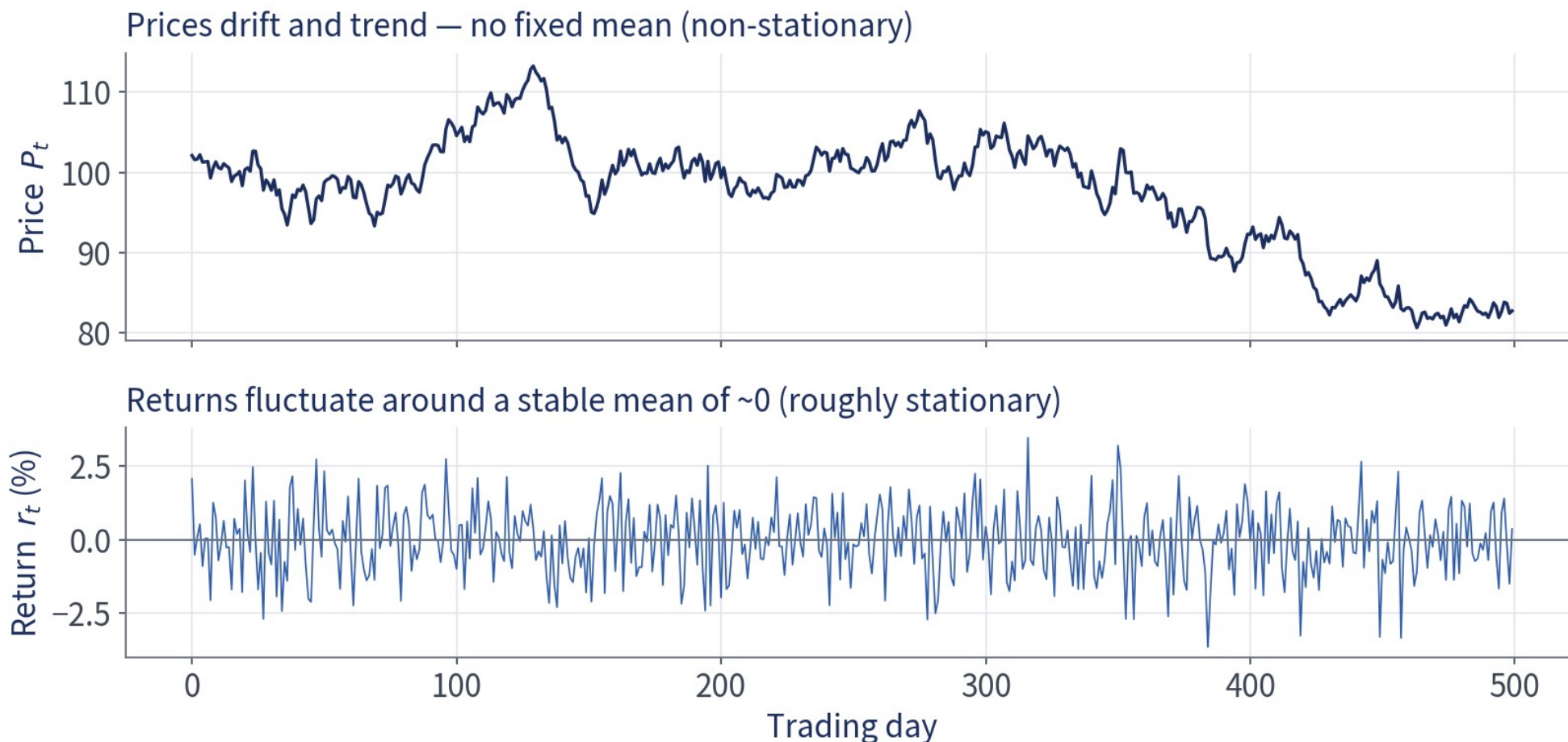
- We never see the true data-generating process.
- We see one sample path and must infer the rules behind it.
- Order matters: yesterday can inform today.

Why this is hard

- Only one history exists — no repeated experiments.
- The rules can change over time (regimes).
- This is why finance leans on stylized facts, not certainty.

Prices Trend; Returns Fluctuate Around Zero

Top: the price wanders with no fixed level. Bottom: its returns hover around a stable mean — this is why we model returns.



Simple (Arithmetic) Return

The intuitive percentage change over one period

$$R_t = \frac{P_t - P_{t-1}}{P_{t-1}} = \frac{P_t}{P_{t-1}} - 1$$

What it answers

- By what percent did my money grow this period?
- Directly interpretable: +3% means +3 cents on the dollar.
- Adds correctly ACROSS assets in a portfolio.

Where it bites

- Does NOT add across time — you must multiply.
- Bounded below by -100% (you cannot lose more than all).
- Right-skewed, so its distribution is awkward to model.

Gross Return and the +1 Convention

The quantity that actually compounds

$$1 + R_t = \frac{P_t}{P_{t-1}} \quad (\text{gross return})$$

Why we add the 1

- Gross return = the growth multiplier of one dollar.
- A +5% return becomes a gross return of 1.05.
- Wealth after k periods = product of gross returns.

A useful habit

- Think in multipliers, not percentages, when compounding.
- $1.05 \times 0.95 = 0.9975 \rightarrow$ a net loss, not break-even.
- This preview is the +20% / -20% trap we meet shortly.

Compounding Simple Returns Is Multiplicative

Over many periods, gross returns multiply — they never simply add

$$1 + R_t(k) = \prod_{j=0}^{k-1} (1 + R_{t-j}) \quad (\text{multiplicative})$$

The correct rule

- Multi-period gross return = product of each period's gross return.
- You cannot sum simple returns to get a multi-period total.

Common mistake

- Adding daily simple returns to get a monthly figure.
- The error grows with volatility and horizon.
- Log returns will fix exactly this problem.

Log (Continuously Compounded) Return

The modeler's default unit

$$r_t = \ln \frac{P_t}{P_{t-1}} = \ln(1 + R_t)$$

Why modelers love it

- Adds cleanly over time (next slide).
- Symmetric and unbounded — friendlier for the normal model.
- The natural unit for continuous-time finance.

The price you pay

- Slightly less intuitive than a plain percentage.
- Does NOT add across assets in a portfolio.
- So convert back to simple for portfolio arithmetic.

Log Returns Add Through Time

The single most convenient property in time-series finance

$$r_t(k) = \sum_{j=0}^{k-1} r_{t-j} \quad (\text{additive through time})$$

Why it works

- $\log(P_T / P_0)$ telescopes into a sum of one-period logs.
- A k -day log return is just the sum of k daily log returns.

What it buys us

- Time-aggregation becomes simple addition.
- Means and variances scale in a clean, tractable way.
- This underpins the annualization rule later in this part.

Simple and Log Returns Are Two Views of One Move

An exact, invertible relationship — not an approximation

$$r_t = \ln(1 + R_t) \quad \Longleftrightarrow \quad R_t = e^{r_t} - 1$$

How to read it

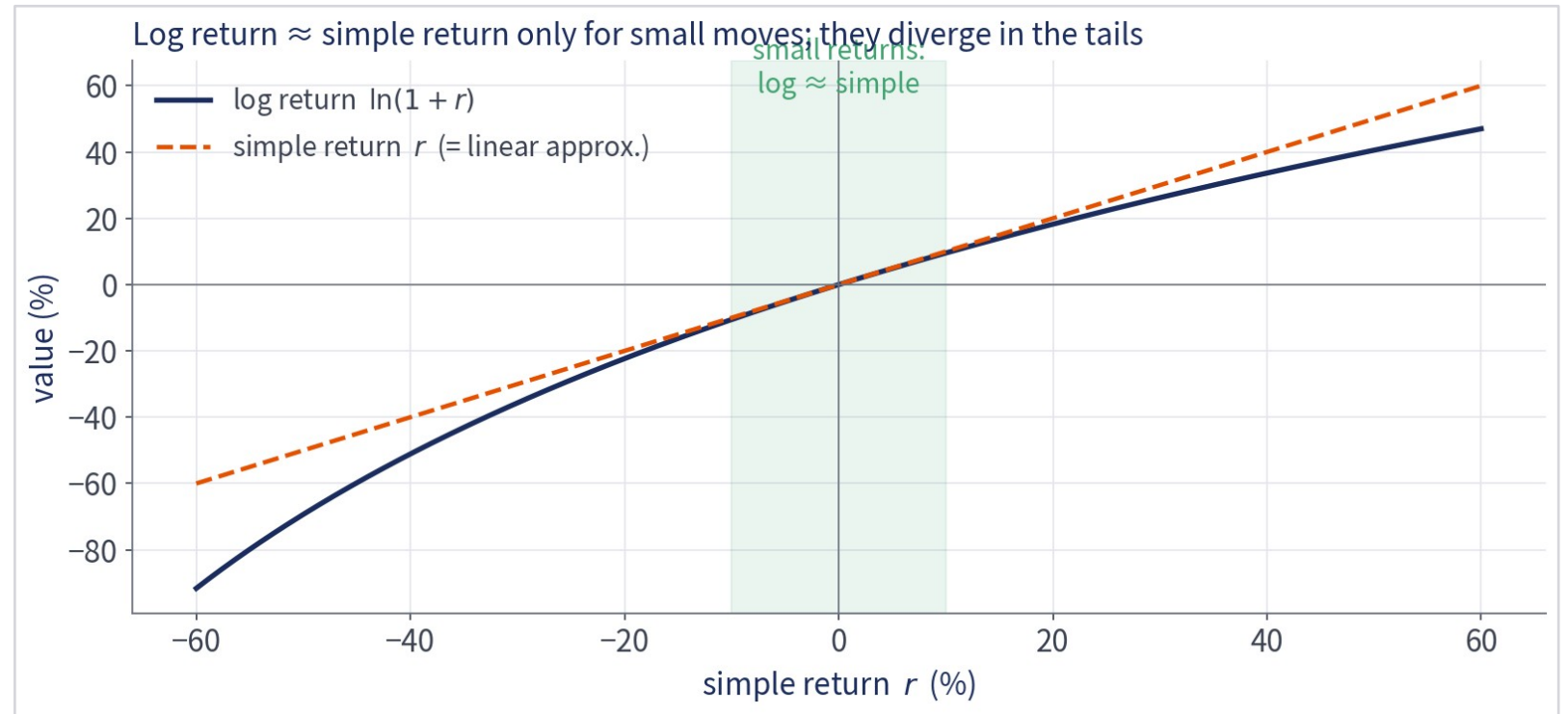
- Given either return, you can recover the other exactly — no information is lost in the conversion.
- Log return is always slightly smaller than the simple return (except at zero), because $\ln(1+R) < R$ for $R > 0$.
- The gap between them is negligible for small moves and grows for large ones — which the next chart makes visual.

When Is Log $\approx$ Simple?

The approximation is excellent for small returns and breaks down in the tails.

Reading the chart

- Green band: $|r| < 10\%$.
- There the two curves overlap.
- Daily equity moves live here.
- Tails: log bends below simple.
- So for crashes, the choice matters.



Computing Both Returns by Hand

A two-day price path — follow every digit

Day	Price P_t	Simple $R_t = \frac{P_t}{P_{t-1}} - 1$	Log $r_t = \ln(P_t/P_{t-1})$
0	100.00	—	—
1	110.00	+10.00%	+9.531%
2	99.00	−10.00%	−10.536%
0→2 total	99.00	−1.00% ($=1.10 \times 0.90 - 1$)	−1.005% ($=+9.531 - 10.536$)

The last row is the point: simple returns multiply ($\rightarrow -1\%$), while log returns add ($\rightarrow -1.005\%$).

Averaging Returns, Way 1: Arithmetic Mean

The simple average of period returns

$$\bar{R}_A = \frac{1}{T} \sum_{t=1}^T R_t \quad (\text{arithmetic mean})$$

What it estimates

- The expected return of a SINGLE future period.
- The right input for forward-looking, one-period models.
- Used in CAPM, mean-variance optimization, forecasts.

The catch

- It overstates the growth you actually realize over time.
- Because it ignores compounding of gains and losses.
- For realized growth, use the geometric mean →

Averaging Returns, Way 2: Geometric Mean

The constant rate that reproduces the actual end wealth

$$R_G = \left[\prod_{t=1}^T (1 + R_t) \right]^{1/T} - 1 \quad (\text{geometric mean})$$

What it estimates

- The realized compound growth rate over the whole path.
- The honest number for 'how did this investment do?'
- Equivalent to the average of log returns.

When to report it

- Historical performance and back-tests.
- Comparing funds over the same past window.
- Never larger than the arithmetic mean (next slide).

Arithmetic $\geq$ Geometric: The Volatility Drag

The two means differ by roughly half the variance

$$R_G \approx \bar{R}_A - \frac{1}{2}\sigma^2 \leq \bar{R}_A$$

The intuition

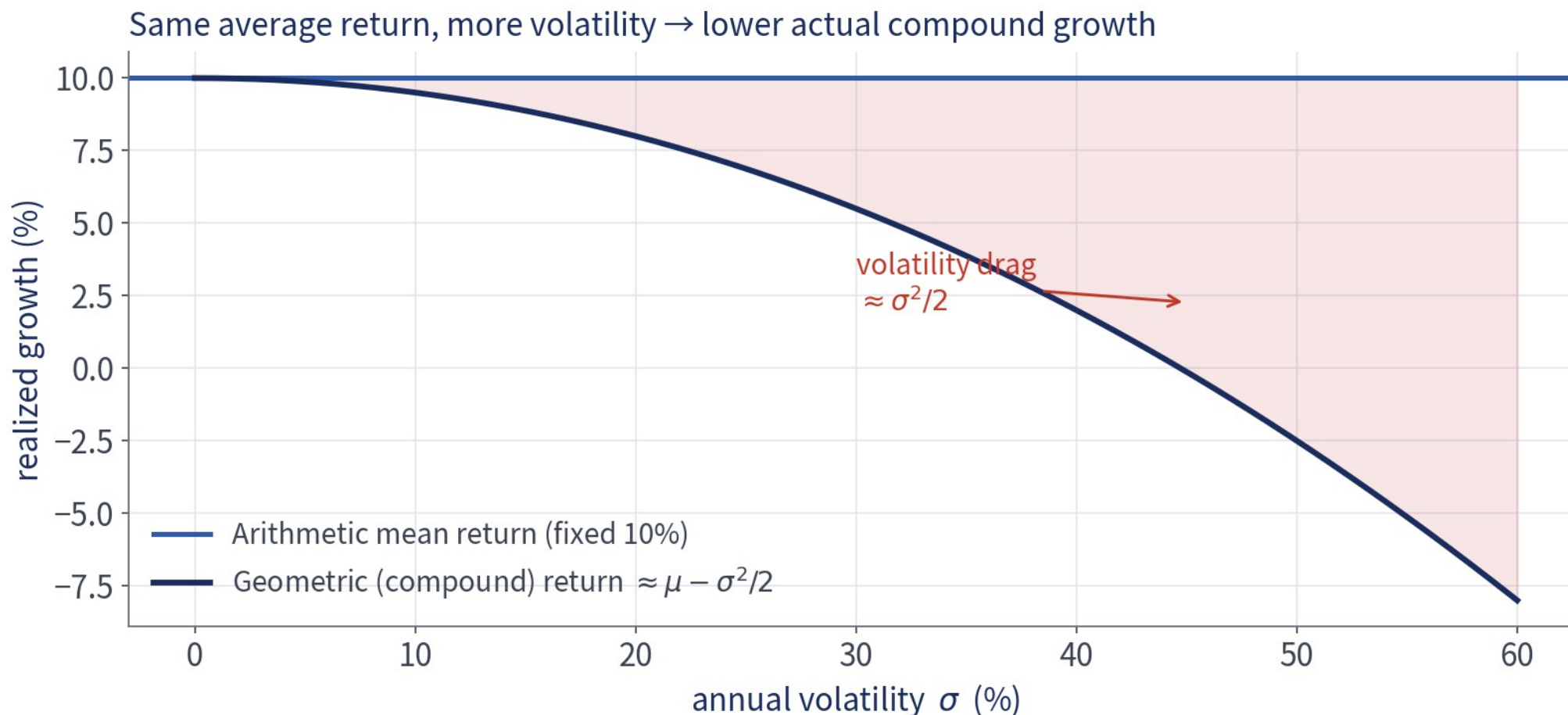
- A -50% loss needs a +100% gain to recover.
- Volatility hurts compound growth asymmetrically.
- More volatility $\rightarrow$ bigger gap between the two means.

The practical warning

- Quoting the arithmetic mean as 'your return' overstates it.
- The drag is about $\sigma^2 / 2$ per period.
- The next chart shows the gap widening with volatility.

Same Average, Less Growth: Volatility Drag

Hold the arithmetic mean fixed at 10%. As volatility rises, the realized compound return falls away by about $\sigma^2/2$.



The +20% / -20% Trap

Why equal-looking gains and losses do not cancel

$$(1 + 0.20)(1 - 0.20) - 1 = 0.96 - 1 = -4\%$$

What just happened

- Up 20% then down 20% leaves you DOWN 4%, not flat.
- Simple returns cannot be added — they compound.
- The loss acts on a larger base than the gain did.

The log-return view

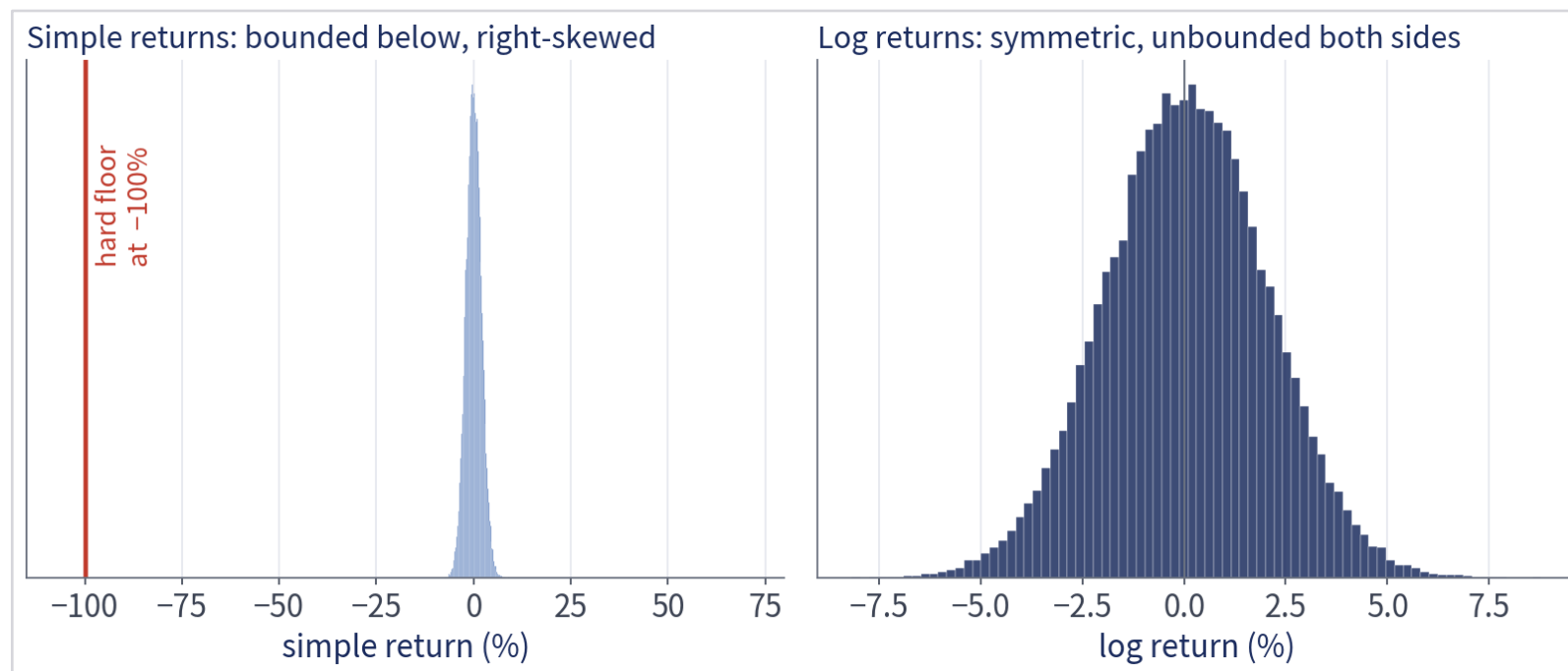
- In logs: +18.23% then -22.31% = -4.08% $\approx \ln(0.96)$.
- Logs add, so the arithmetic is transparent.
- This is exactly why we prefer logs over time.

Why Log Is the Modeling Default

The distributional shapes differ — and one is far friendlier to standard statistics.

What to notice

- Simple: floor at -100% .
- So it is right-skewed.
- Log: symmetric, no floor.
- Closer to a normal shape.
- Hence the modeling default.

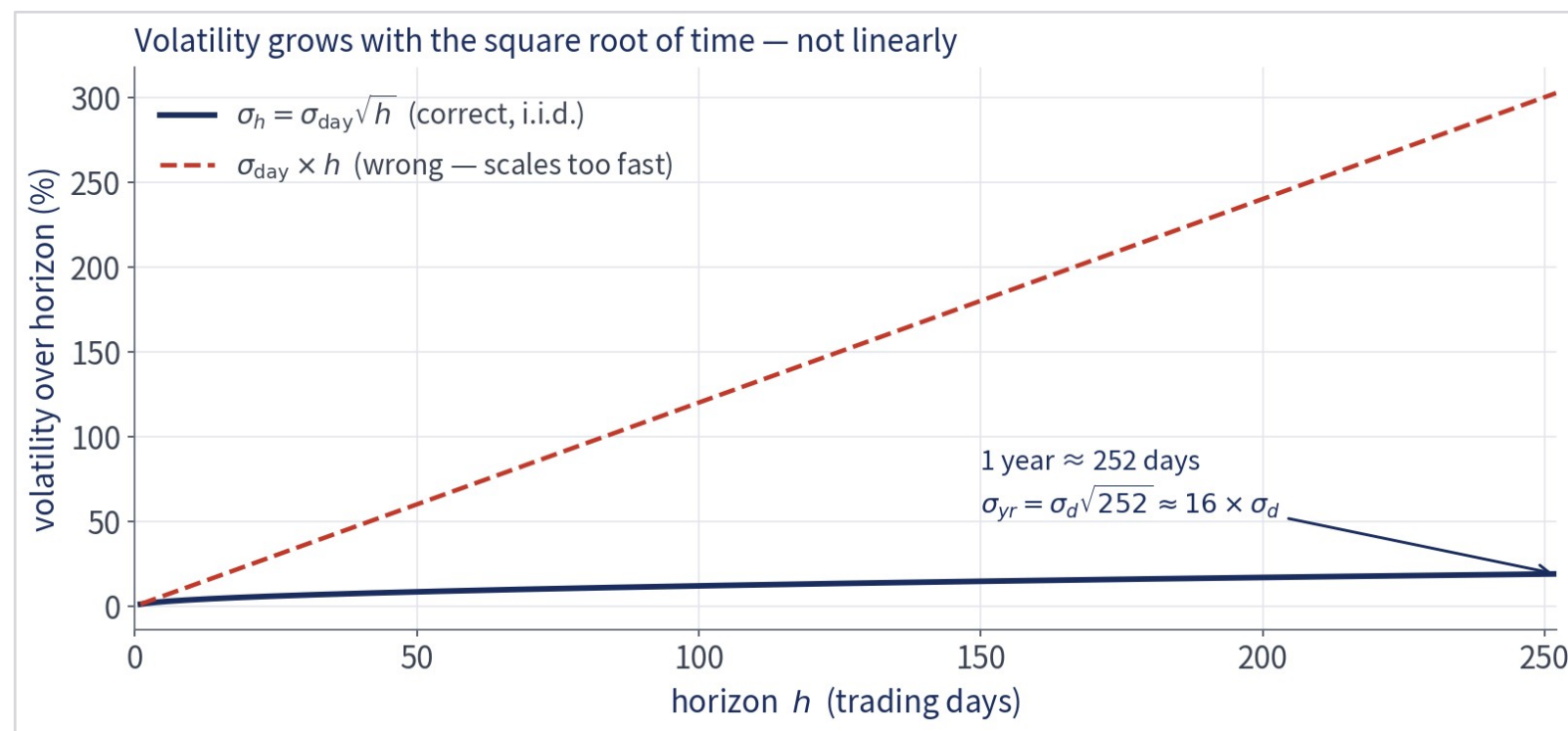


Annualizing: The Square-Root-of-Time Rule

Under i.i.d. returns, the mean scales with time but volatility scales with its square root.

The rule

- Mean: multiply by h .
- Vol: multiply by $\sqrt{h}$.
- 1 year ≈ 252 days.
- So $\sigma_{yr} \approx 16 \times \sigma_{day}$.
- Assumes independence.



Real, Nominal, and Excess Returns

Three adjustments you must make before comparing returns fairly

$$1 + R_t^{\text{real}} = \frac{1 + R_t^{\text{nom}}}{1 + \pi_t}, \quad R_t^e = R_t - R_t^f$$

Real vs nominal

- Nominal returns ignore inflation.
- Real return strips out the change in the price level.
- Compare purchasing power, not headline numbers.

Excess return

- Subtract the risk-free rate to isolate the risk premium.
- This is what asset-pricing models actually explain.
- Also remember total return: add reinvested dividends.

Which Return Should You Use?

Across assets → simple. Across time → log. For an average: arithmetic looks forward, geometric looks back.

Which return should you use?

The answer depends on whether you are adding across assets or across time.

Adding ACROSS assets (a portfolio at one date)

Use SIMPLE returns

Portfolio return = Sum of (weight x simple return).
Log returns are NOT additive across assets.

Adding OVER time (many periods compounding)

Use LOG returns

T-period log return = Sum of one-period log returns.
They add up cleanly through time.

Summarizing "the average return" over a path:

- **Arithmetic mean** = best guess for the NEXT single period (forward-looking).
- **Geometric mean** = the REALIZED compound growth you actually earned (backward-looking).

Six Ways Returns Go Wrong in Practice

The traps that separate a clean analysis from a misleading one

Pitfall	What goes wrong	The fix
Adding simple returns over time	Distorts the multi-period total	Multiply gross returns (or use logs)
Quoting the arithmetic mean as return	Ignores vol drag; overstates growth	Use the geometric mean instead
Ignoring dividends	Understates the true total return	Use total-return (adjusted) prices
Annualizing volatility linearly	Scales too fast; inflates risk	Scale by $\sqrt{\text{time}}$, not time
Mixing currencies	Confounds asset return with FX	Convert consistently; state the base
Survivorship in the data	Dead names dropped; looks too good	Use survivorship-free, point-in-time data

From Prices to Returns

No manual coding — you direct Claude Code and read what it writes

GOAL Build a reusable returns pipeline by prompting Claude Code, then read and verify the code it produces.

[INPUT] Prompt · Claude Code

In this repo, create `returns.py` that:

1. downloads daily ADJUSTED prices for a ticker I pass (default AAPL) via `yfinance`, from 2015-01-01 to today;
2. computes simple and log returns;
3. verifies $r_{\text{log}} = \ln(1 + R_{\text{simple}})$ per row;
4. checks that summed monthly log returns equal $\ln(\text{product of gross returns})$;
5. saves `returns.csv` and plots the price, both return series, and their histograms.

Then run it on AAPL and show me the head of `returns.csv` plus the saved figures.

How to work with it

- Read the code — you must understand every line.
- Change the ticker and dates, then re-run.
- If a check fails, tell Claude what you expected.
- Keep `returns.csv` — Labs 2 and 3 reuse it.

Deliverable: `returns.py` + `returns.csv` + the figures, reused across the whole week.

Three Questions Before You Move On

If you can answer these, Part 1 has landed

Q1

A stock returns +20% then -20% (simple). What is the two-day total?

-4%, not 0%

Q2

You want to aggregate one asset's returns over 12 months. Simple or log?

Log (additive)

Q3

Which mean should a fund report for its realized 10-year track record?

Geometric

Cover the right column first, answer, then check. All three follow directly from this part.

Three Things to Carry Forward

Part 1 in one screen

1

Model returns, and default to log returns

Prices are non-stationary; returns are comparable and tractable. Log returns add through time and behave better statistically.

2

Match the mean to the question

Arithmetic mean for the expected next period; geometric mean for realized growth. They differ by the volatility drag, $\approx \sigma^2/2$.

3

Respect the traps

Don't add simple returns over time, don't annualize vol linearly, and always use total-return, survivorship-free data.

Next — Part 2: the Stylized Facts of Returns, the empirical patterns every model must explain.

★ WEEK 1 · PART 1 · PRACTICE

Returns, Worked by Hand

Simple and Log Returns, and Why Gains Don't Cancel Losses

Two views of one move, and the volatility drag

Part 1's foundations, with every digit shown. We compute simple and log returns from a price path - watching log returns add while simple returns multiply - then confront the volatility drag, where the arithmetic mean flatters and the geometric mean tells the truth.

1P

PART 1 · APPLICATIONS

The Unit Everything Is Measured In

Case A: compute simple and log returns from a two-day price path, and see why +10% then -10% leaves you down.
Case B: the arithmetic mean says break-even while the geometric mean shows a loss - the volatility drag, equal to about half the variance. Then a Claude Code lab.

What Part 1 Told Us

Now we apply it - with numbers

1

We model returns, not prices

Returns are comparable and better behaved than raw prices. Simple returns compound multiplicatively; log returns add through time.

2

Simple and log are two views of one move

An exact, invertible relationship: the log return is the log of one plus the simple return. Close for small moves, not identical.

3

The geometric mean tells the truth

The arithmetic mean overstates compound growth. The gap - the volatility drag - is about half the variance.

Everything here is a calculator exercise: one returns table, one volatility drag.

How These Cases Work

Data, then arithmetic, then meaning

Each case has three steps

- Start from a price path or a pair of returns.
- Compute both kinds, or both means.
- Then read what really happened.

What you should watch

- How log returns add and simple returns multiply.
- Why equal gains and losses don't cancel.
- How the drag equals half the variance.

Simple and Log Returns

Two views of one price path

A stock goes from 100 to 110, then to 99. We compute both the simple and the log return each day, and watch two facts emerge: log returns add up to the total log return, while gross simple returns multiply - and a +10% followed by a -10% leaves you at 99, not 100. Every digit is shown.

The price path

- Starts at 100.
- Up to 110, a 10% gain.
- Down to 99, a 10% loss.

What we'll do by hand

- Compute simple and log returns each day.
- Add the logs; multiply the gross simples.
- See where you actually end up.

The Two Definitions

Simple, gross, and log

$$R_t = \frac{P_t}{P_{t-1}} - 1 \quad (\text{simple}), \quad r_t = \ln \frac{P_t}{P_{t-1}} = \ln(1 + R_t) \quad (\text{log})$$

The simple return

- The percentage change over the period.
- Intuitive, and what most people quote.
- It compounds by multiplying.

The log return

- The log of the gross return.
- The modeler's default unit.
- It compounds by adding.

Both Returns, Day by Day

Follow every digit

day	price	simple R	log r
start	100	-	-
day 1	110	+10.00%	+9.53%
day 2	99	-10.00%	-10.54%
total	99	-1.00%	-1.005%

Day one is a clean +10%; day two is -10% off the higher base of 110, which costs 11 points. The simple returns don't sum to the total, but the log returns do: 9.53% minus 10.54% is -1.005%, exactly the log of 99 over 100.

Logs Add, Simples Multiply

Why +10% and -10% don't cancel

$$\ln(1.10) + \ln(0.90) = 0.0953 - 0.1054 = -0.01005 = \ln \frac{99}{100} \quad (\text{logs add; } +10\%, -10\% \rightarrow 99)$$

The log side

- The two log returns simply add.
- 9.53% minus 10.54% is -1.005%.
- That equals the total log return.

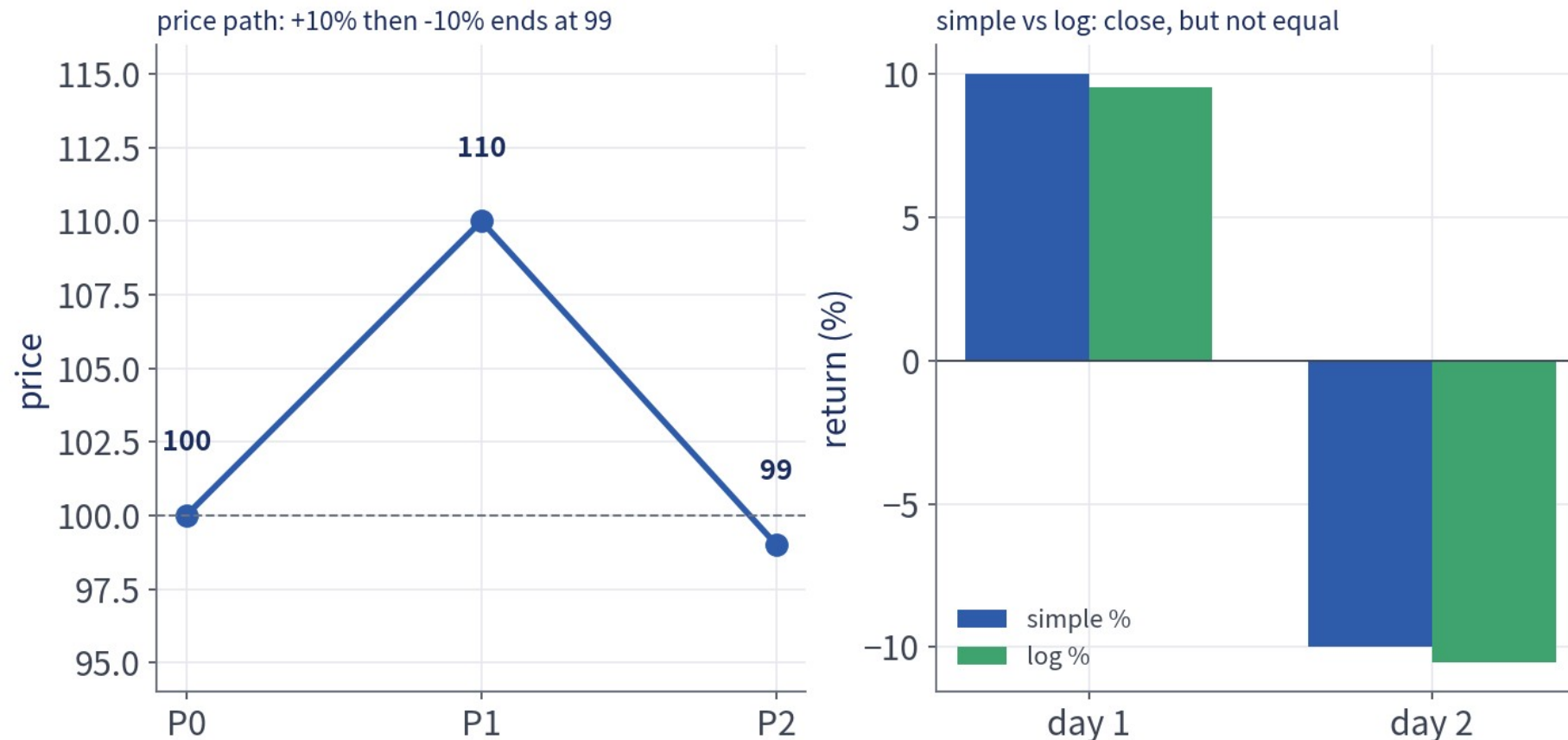
The simple side

- The gross returns multiply: 1.1 times 0.9.
- That's 0.99 - a total of -1%.
- You end at 99, not back at 100.

Two Views, One Move

Left, the price path: up 10% to 110, then down 10% off that higher base, landing at 99 - below where it started. Right, the daily returns: simple and log sit close together for these small moves but are never exactly equal, the log always a touch smaller on the way up and larger in magnitude on the way down.

Case A: two views of the same two moves



Why Log Is the Default

The modeler's convenience

Why modelers prefer logs

- They add across time, so multi-period is easy.
- They're symmetric around zero.
- They pair naturally with the normal distribution.

When simple is right

- For actual portfolio percentage changes.
- For quoting a client's realized gain.
- The two agree closely for small moves.

The Volatility Drag

When the average lies

Consider a return of +20% followed by -20%. The arithmetic mean is exactly zero, suggesting break-even - but you actually end at 96, down 4%. The geometric mean captures this: it's negative. We compute both, measure the gap - the volatility drag - and confirm it equals about half the variance.

The two returns

- Up 20%, then down 20%.
- Arithmetic mean: exactly zero.
- But the wealth path disagrees.

What we'll do by hand

- Compute the geometric mean.
- Measure the drag against the arithmetic.
- Match it to half the variance.

Arithmetic vs Geometric

Two ways to average returns

$$\bar{R}_{\text{arith}} = \frac{1}{n} \sum R_t, \quad \bar{R}_{\text{geo}} = \left(\prod (1 + R_t) \right)^{1/n} - 1 \leq \bar{R}_{\text{arith}}$$

The arithmetic mean

- Just the simple average of the returns.
- Answers: what to expect next period.
- But it overstates realized growth.

The geometric mean

- The constant rate that reproduces end wealth.
- Always less than or equal to the arithmetic.
- It's the honest compound rate.

The +20% / -20% Trap

Zero average, real loss

$$+20\%, -20\% : \bar{R}_{\text{arith}} = 0, \text{ but } 1.2 \times 0.8 = 0.96 \Rightarrow \bar{R}_{\text{geo}} = -\mathbf{2.02\%}$$

The arithmetic view

- Plus 20 and minus 20 average to zero.
- It looks like you broke even.
- You did not.

The geometric view

- 1.2 times 0.8 is 0.96 - down 4%.
- Per period, that's minus 2.02%.
- The wealth path ends at 96.

The Drag Is Half the Variance

And real vs nominal vs excess

$$\text{drag} = \bar{R}_{\text{arith}} - \bar{R}_{\text{geo}} = 0.0202 \approx \frac{1}{2}\sigma^2 = 0.02 \quad (\bar{R}_{\text{geo}} \approx \bar{R}_{\text{arith}} - \frac{1}{2}\sigma^2)$$

The drag

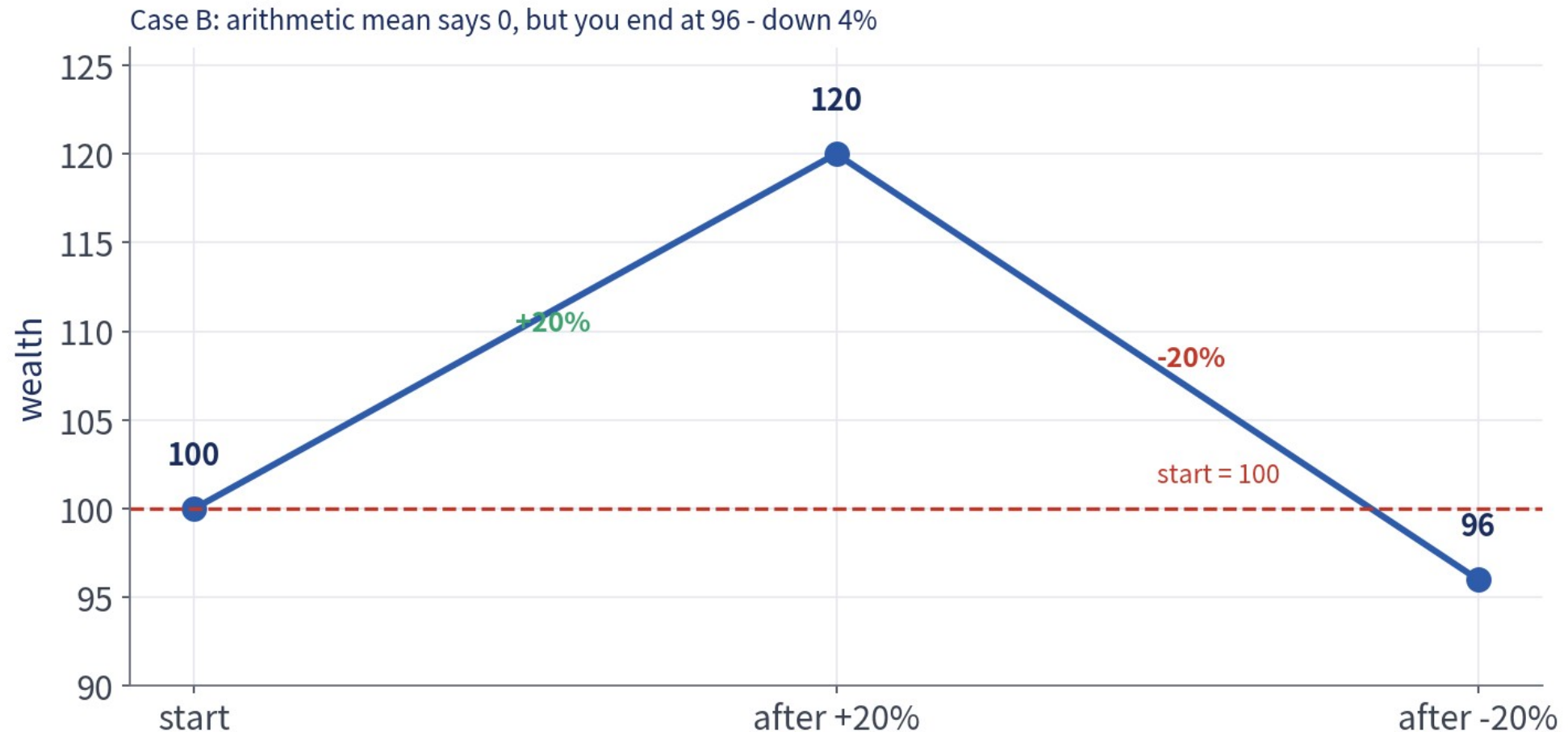
- Arithmetic minus geometric is 0.0202.
- Half the variance is 0.02.
- They match - drag is about half the variance.

Three adjustments

- Real: strip out inflation - 8% over 3% is 4.85%.
- Excess: subtract the risk-free - 8% minus 2% is 6%.
- Compare returns only after these.

Zero on Average, Down in Reality

Start at 100, gain 20% to 120, then lose 20% off that higher base - and land at 96, not 100. The arithmetic mean of the two returns is zero, but the wealth path tells the truth: a 4% loss. This gap between the average return and the realized growth is the volatility drag.



Comparing Returns Fairly

The adjustments that matter

Why the drag matters

- Volatile strategies lose to the drag.
- Quote the geometric mean for realized growth.
- The wider the swings, the bigger the gap.

Real, nominal, excess

- Nominal returns ignore inflation.
- Real returns strip it out - the true purchasing power.
- Excess returns subtract the risk-free rate.

The Returns Toolkit

The unit everything is built on

Returns are the foundation of everything that follows. Convert prices to returns, choosing log returns when you'll model and sum across time, simple returns when you'll compound a portfolio. Average with the geometric mean for honest realized growth, remembering the volatility drag of half the variance. And adjust for inflation and the risk-free rate before comparing. Get the unit right, and every later model inherits a clean input.

The returns (Case A)

- Simple +10%, -10%; log +9.53%, -10.54%.
- Logs add to -1.005%; simples multiply to -1%.
- End at 99, not 100.

The drag (Case B)

- Arithmetic mean 0, geometric -2.02%.
- Drag 0.0202, about half the variance.
- The average lied; the wealth path didn't.

Reproduce the Returns and the Drag

Direct Claude Code to check your arithmetic

GOAL Compute both returns from a price path, then the arithmetic and geometric means.

[INPUT] Prompt · Claude Code

Write and run `part1_practice.py` that:

CASE A - returns:

- * for prices `[100, 110, 99]`, print the simple and log return each day;
- * confirm the logs sum to $\ln(99/100)$ and the gross simples multiply to 0.99.

CASE B - the drag:

- * for returns `[+0.20, -0.20]`, print the arithmetic and geometric mean;
- * confirm the drag is about half the variance.

Then plot the price path and the +20/-20 wealth path.

Check against the slides

- logs: +9.53%, -10.54% -> -1.005%.
- gross: $1.1 \times 0.9 = 0.99$.
- arith 0, geo -2.02%.
- drag 0.0202 ~ 0.02.

Deliverable: `part1_practice.py` whose printed numbers match the returns table and the drag.

Three Questions on the Cases

If you can answer these, the practice has landed

Q1 How do log returns compound over time?

They add; gross simple returns multiply

Q2 Why don't +10% and -10% cancel?

The loss is off a higher base - you end at 99

Q3 How big is the volatility drag?

About half the variance of the returns

All three come from the returns table, the multiplicative compounding, and the volatility drag.

Three Pitfalls to Avoid

Where returns trip people

Common mistakes

- "A +10% then -10% gets you back to even."
- "The arithmetic mean is the realized growth."
- "Log and simple returns are the same."

The reality

- The loss is off a higher base - you end down.
- Use the geometric mean; mind the drag.
- They agree only for small moves - not exactly.

Two Calculations to Remember

Part 1 applications in one screen

1

Log returns add; gains don't cancel losses

On the path 100 to 110 to 99, the log returns of +9.53% and -10.54% summed to -1.005%, exactly the log of 99 over 100, while the gross simples multiplied to 0.99. A +10% then -10% ends at 99 - the loss falls on a higher base.

2

The geometric mean tells the truth

For +20% then -20%, the arithmetic mean was zero but the wealth ended at 96 - the geometric mean was -2.02%. The gap, 0.0202, matched half the variance of 0.02: the volatility drag, geometric equals arithmetic minus half the variance.

3

Adjust before you compare

Real returns strip out inflation - 8% over 3% is 4.85%, not 5%; excess returns subtract the risk-free rate. Choose log returns to model and sum, simple returns to compound a portfolio, and always average with the geometric mean.

Next — Part 2 practice: the stylized facts of returns, from fat tails to volatility clustering.

★ WEEK 1 · PART 2 of 3

The Stylized Facts of Returns

The Empirical Fingerprints of Financial Markets

Fat tails · volatility clustering · leverage · the predictability split

Across nearly every market and era, returns share a handful of robust statistical patterns. These 'stylized facts' are the targets every model in this course must reproduce.

2

PART 2 · STYLIZED FACTS

What Every Model Must Explain

Before we can model returns, we must know what real returns actually look like. This part catalogs the six patterns that show up everywhere.

Why Study Stylized Facts First?

Three reasons they come before any equation

1

They are the benchmark every model is judged against

A model is only useful if it reproduces what we see in the data. The stylized facts define 'what we see'.

2

They are astonishingly universal

The same patterns appear in stocks, bonds, FX, and crypto — across centuries and continents. That universality is itself a clue.

3

They tell us where prediction is possible — and where it is not

The single most important fact splits the world: the direction of returns is unpredictable, but their magnitude is not.

A model that cannot reproduce the stylized facts is wrong, no matter how elegant.

What Is a Stylized Fact?

A robust empirical regularity, not a theorem

A stylized fact is a statistical pattern that reappears across almost all financial return series — a qualitative truth robust enough to build models around.

What qualifies

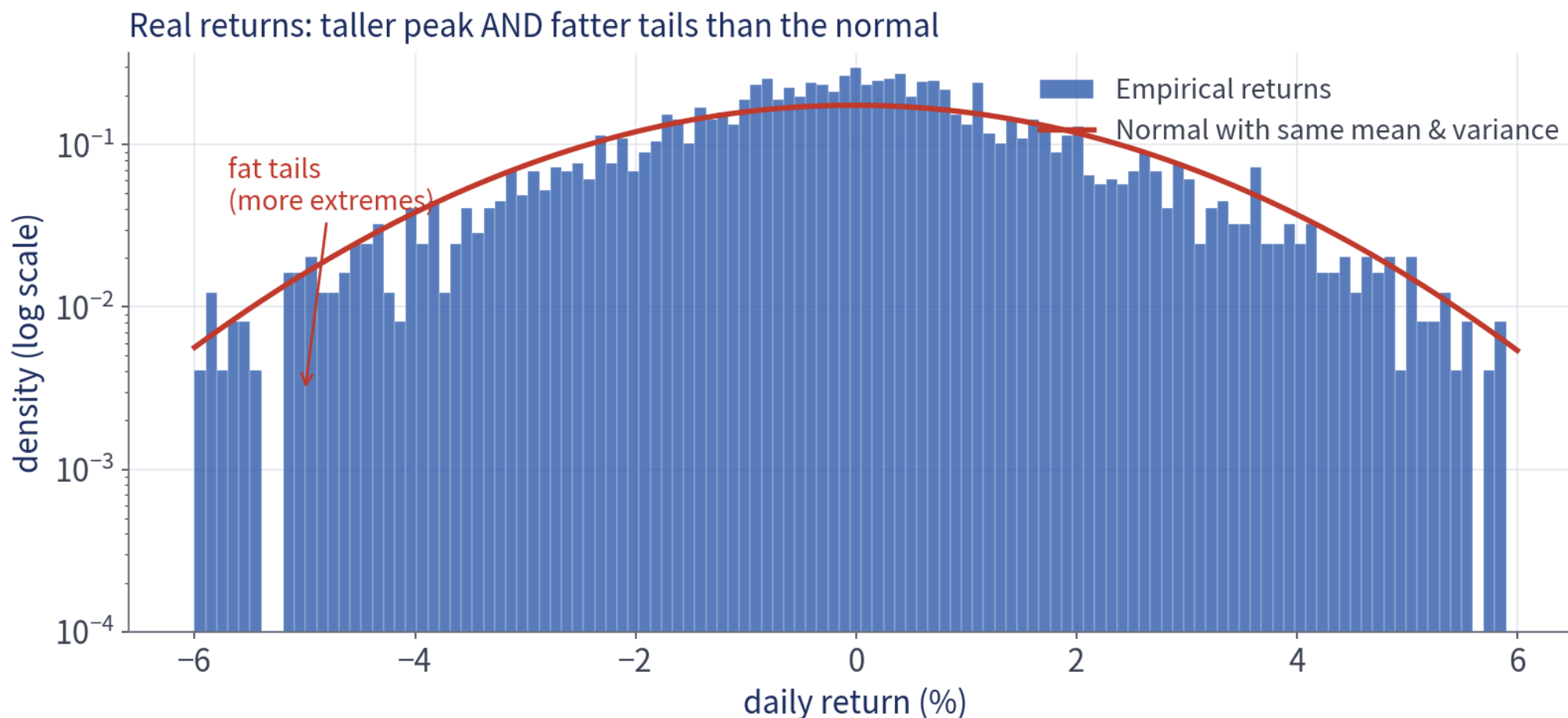
- Seen across many assets, markets, and time periods.
- Survives changes in sample and frequency.
- Qualitative and robust — not a precise numerical law.

How we use them

- As a checklist: does my model reproduce them?
- To rule out models that assume normal, i.i.d. returns.
- To motivate the tools of Weeks 2 onward (GARCH, etc.).

The Distribution of Returns Is Not Normal

Compared with a normal of the same mean and variance, real returns have a taller central peak and much fatter tails.



Describing a Distribution, Part 1: Mean & Variance

The first two moments — center and spread

$$\mu = \mathbb{E}[R], \quad \sigma^2 = \mathbb{E}[(R - \mu)^2], \quad \sigma = \sqrt{\sigma^2}$$

Mean μ — the center

- The average return; where the distribution sits.
- For daily equity returns it is tiny and hard to estimate.

Variance σ^2 / vol σ — the spread

- How widely returns scatter around the mean.
- Volatility σ is the everyday measure of risk.
- Much easier to estimate than the mean — and it moves.

Describing a Distribution, Part 2: Skewness & Kurtosis

The third and fourth moments — asymmetry and tail weight

$$S = \frac{\mathbb{E}[(R-\mu)^3]}{\sigma^3}, \quad K = \frac{\mathbb{E}[(R-\mu)^4]}{\sigma^4}$$

Skewness S — the lean

- $S < 0$: a longer left tail (big losses more likely).
- $S > 0$: a longer right tail.
- Equity returns are typically negatively skewed.

Kurtosis K — the tails

- Measures how much probability sits in the tails.
- Normal has $K = 3$; anything above means fat tails.
- Excess kurtosis = $K - 3$ is the fat-tail gauge.

The Autocorrelation Function (ACF)

Our tool for asking: does the past predict the future?

$$\rho_R(k) = \frac{\text{Cov}(R_t, R_{t-k})}{\text{Var}(R_t)}$$

What it measures

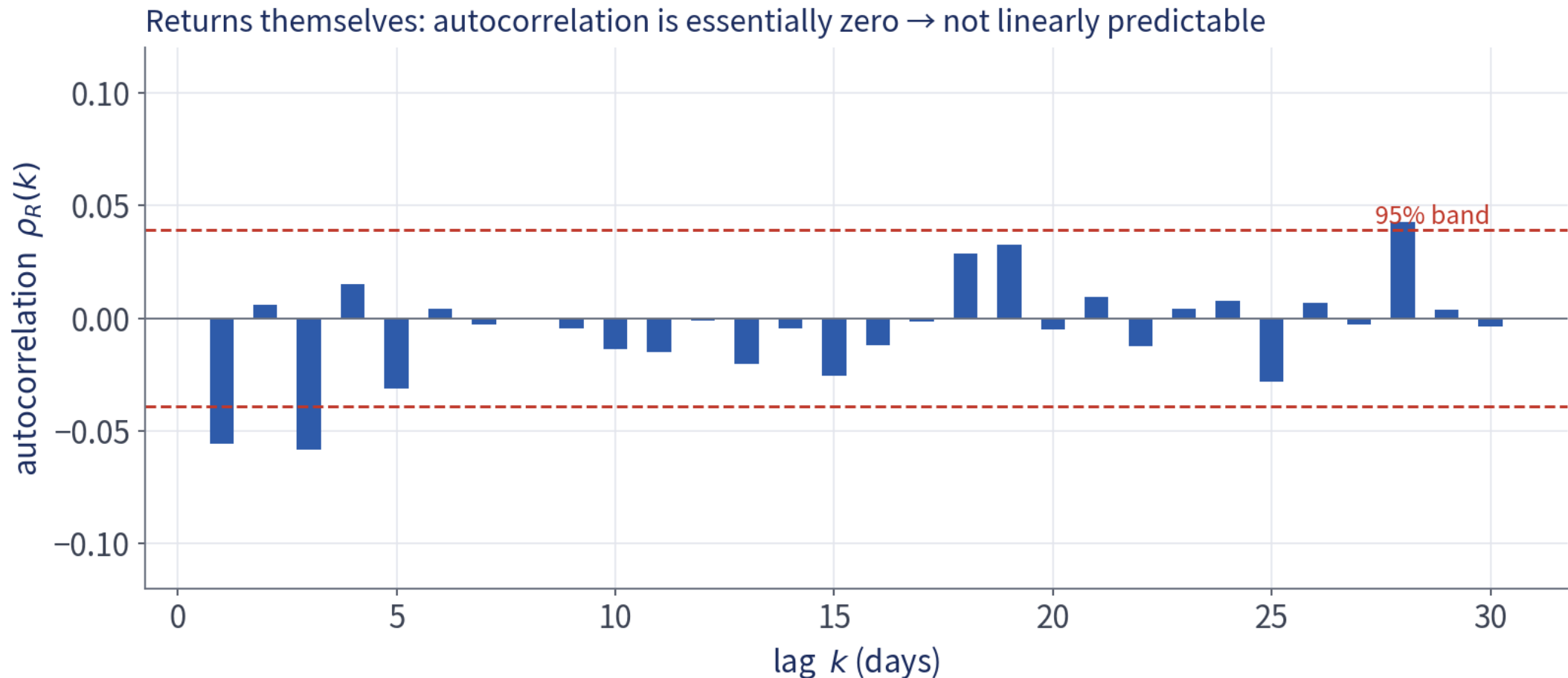
- The correlation of a series with its own past, k steps back.
- $\rho(k)$ near 0 $\rightarrow$ no linear link between today and lag k .
- $\rho(k)$ far from 0 $\rightarrow$ predictable structure at that lag.

How to read the plot

- Bars are $\rho(k)$ at each lag $k = 1, 2, 3, \dots$
- Dashed lines are the 95% 'no-correlation' band.
- Bars inside the band are statistically indistinguishable from 0.

Returns Have Almost No Autocorrelation

Every bar sits inside the 95% band: $\rho_R(k) \approx 0$. The direction of returns is not linearly predictable — the basis of market efficiency (Part 3).



Fat Tails (Excess Kurtosis)

Rare does not mean impossible

$$\text{excess kurtosis} = K - 3 > 0 \implies \text{fat tails}$$

What it means

- Extreme moves happen far more often than a normal predicts.
- A 'once-in-a-century' normal event can arrive every few years.
- This return series has excess kurtosis of about 5.

Why it matters

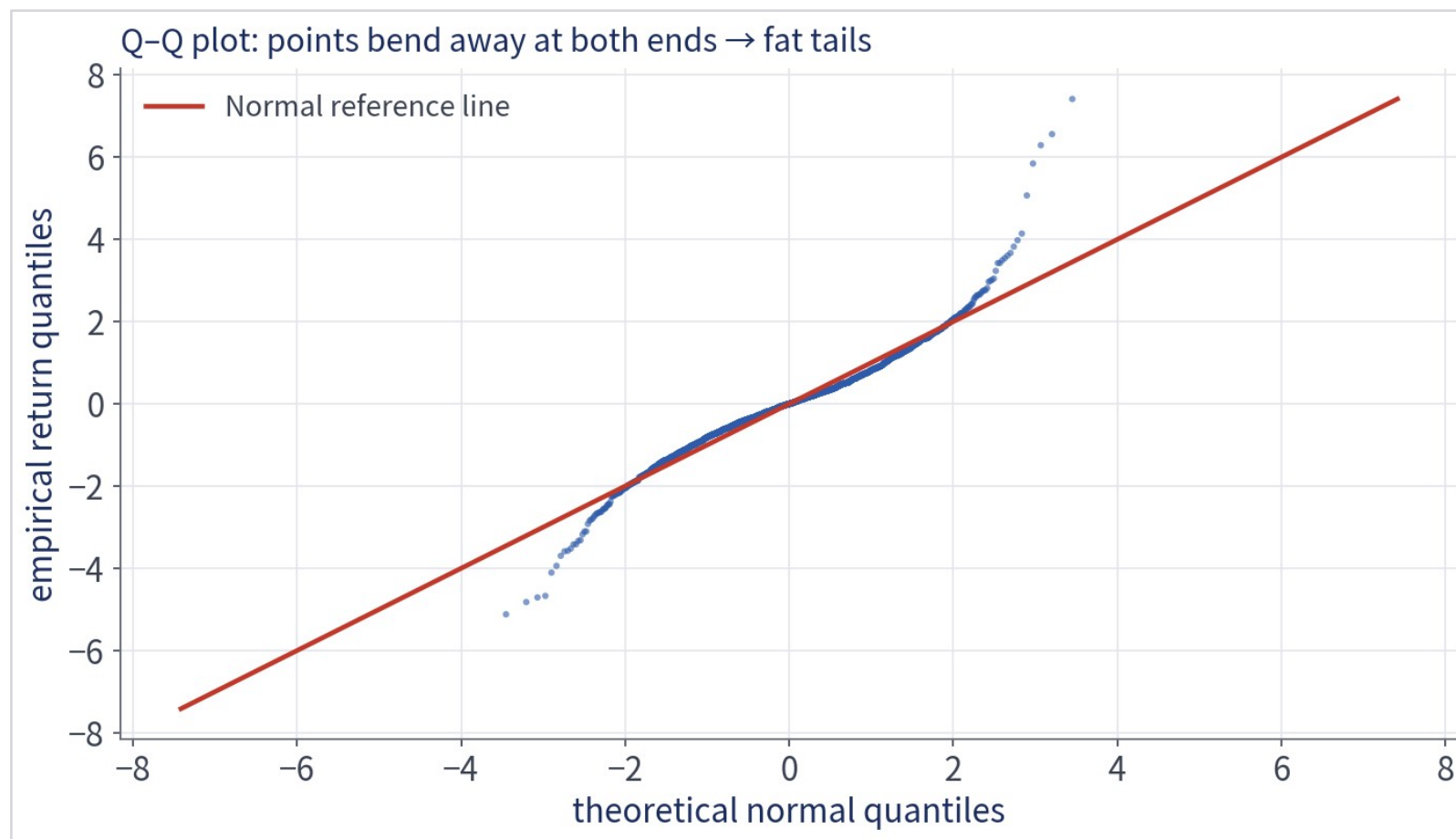
- Risk models built on the normal understate tail risk badly.
- Value-at-Risk and stress tests must use fat-tailed models.
- The next chart makes the fat tails visible.

Seeing Fat Tails: The Q-Q Plot

If returns were normal, the points would lie on the red line.

Reading a Q-Q plot

- Points on line = normal.
- Both ends bend away.
- Low end below the line.
- High end above the line.
- That S-shape = fat tails.



How Heavy Are the Tails?

From a single number to a tail law

Beyond kurtosis, we describe tails by a power law: the probability of an extreme move of size x decays like x raised to a negative tail index α .

The tail index α

- Smaller $\alpha \rightarrow$ heavier tails, more extreme events.
- Equity returns typically show α of roughly 3 to 5.
- Only moments up to order α are finite.

The Student-t link

- A Student-t with ν degrees of freedom has tail index ν .
- Small ν (e.g. 4–6) reproduces realistic fat tails.
- As ν grows, the t converges back to the normal.

Distributions That Capture Fat Tails

Choosing the right model for the tails

Distribution	Tail behavior	Key parameter	Use when
Normal (Gaussian)	Thin — decays very fast	mean, variance	A first approximation, or long horizons
Student-t	Fat — power law	degrees of freedom ν	Modeling realistic daily fat tails
Generalized error / Laplace	Heavier than normal	shape parameter	Peaked center with moderate tails
Empirical / EVT	Data-driven, tail index	threshold, tail index	Estimating extreme quantiles (VaR)

Rule of thumb: start normal for intuition, switch to Student-t (or EVT for the far tail) for real risk work.

Gain / Loss Asymmetry (Negative Skew)

Markets fall faster than they rise

Equity return distributions typically lean left: the left tail is longer than the right. Large losses are more likely than large gains of the same size.

What you observe

- Sharp, deep crashes; slower, steadier climbs.
- The biggest single-day moves are usually down days.
- Skewness S is negative for most equity indices.

Why it happens

- Fear spreads and forced selling feed on itself.
- Deleveraging and margin calls amplify declines.
- Closely tied to the leverage effect (a few slides on).

Volatility Clustering

Turbulence comes in waves

"Large changes tend to be followed by large changes, of either sign, and small changes by small changes." — Mandelbrot, 1963

The signature

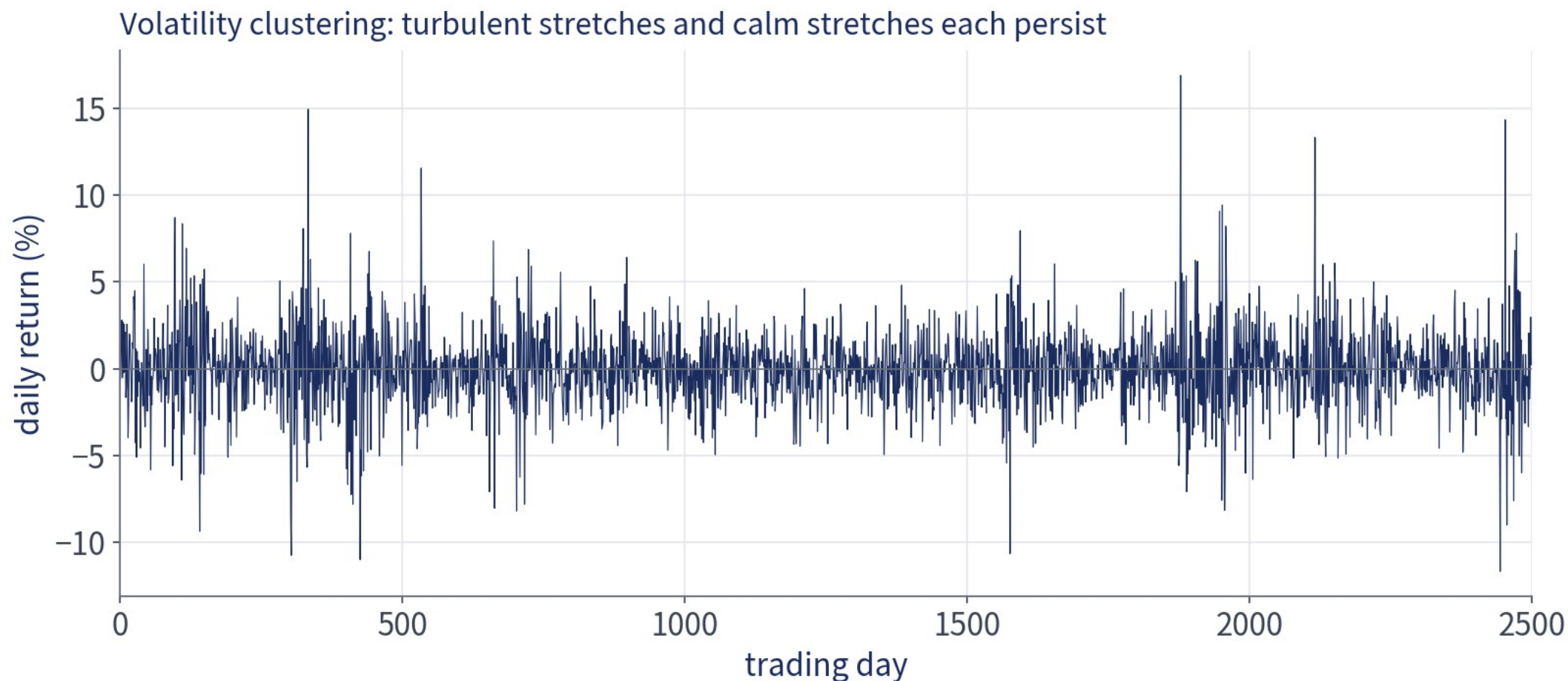
- Calm periods and stormy periods each persist.
- Volatility today is a strong predictor of volatility tomorrow.
- This is what makes risk forecasting possible.

The precise statement

- Returns themselves show no autocorrelation ...
- ... yet their squares (or absolute values) do.
- Two charts make this split unmistakable.

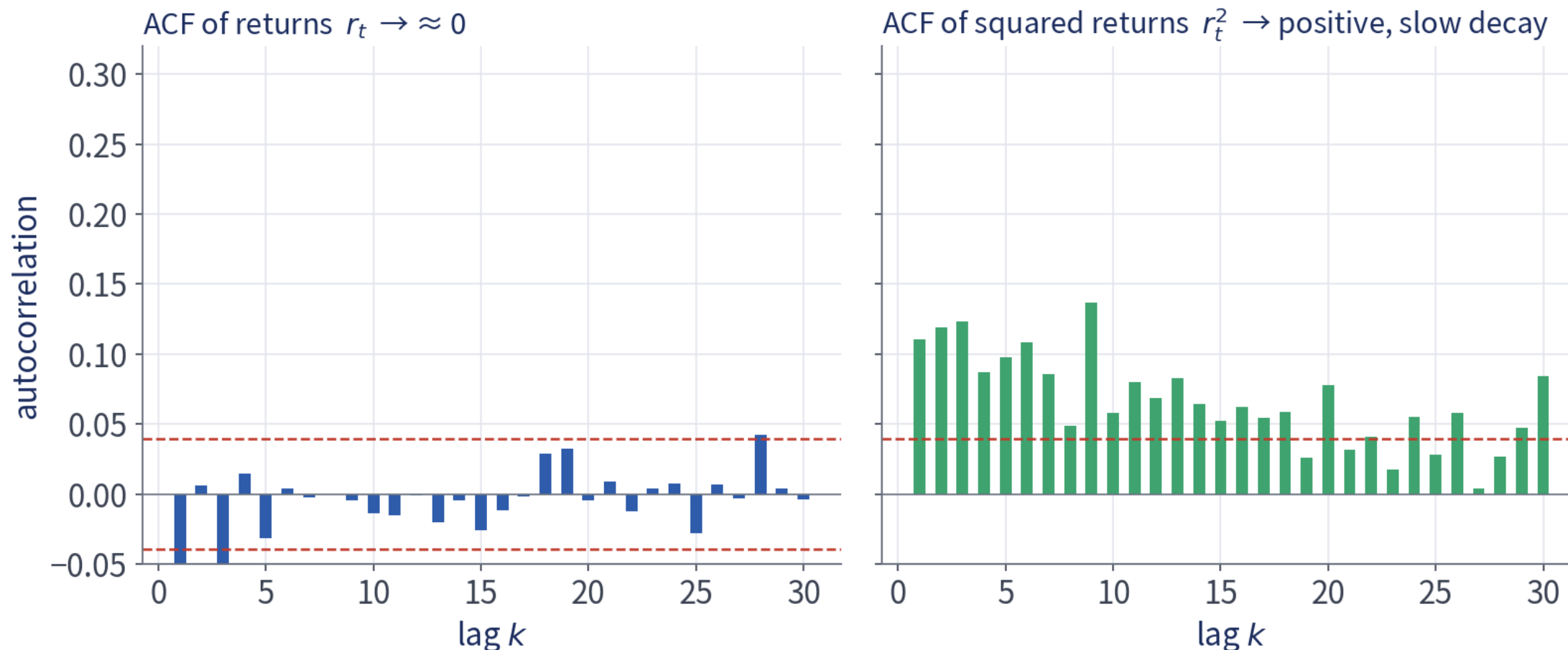
Volatility Clustering, Made Visible

Notice the alternating regimes: stretches of small wiggles, then bursts of large swings. This structure is absent from i.i.d. normal data.



The Split: Returns vs Squared Returns

Left: returns are uncorrelated. Right: squared returns are strongly, slowly-decaying autocorrelated — volatility has long memory.



Returns Are Unpredictable; Volatility Is Not

The single most important split in the course

$$\underbrace{\rho_R(k) \approx 0}_{\text{returns: no memory}}$$

$$\underbrace{\rho_{R^2}(k) > 0, \text{ slow decay}}_{\text{volatility: long memory}}$$

The level (returns)

- No linear memory — you cannot forecast the sign.
- This is the empirical face of market efficiency.
- Direction bets have no statistical edge from the past.

The magnitude (volatility)

- Strong, persistent memory — you CAN forecast risk.
- This is what GARCH-type models exploit (Week 2+).
- Risk management lives entirely on this side.

The Leverage Effect

Bad news bites harder than good news

$$\text{Corr}(r_t, \sigma_{t+1}^2) < 0 \quad (\text{leverage effect})$$

What it says

- A negative return raises future volatility more than a positive one of equal size.
- So returns and future volatility are negatively correlated.

Why the name

- A price drop raises a firm's debt-to-equity ratio ...
- ... making its equity riskier, hence more volatile.
- Behavioral fear effects reinforce the same pattern.

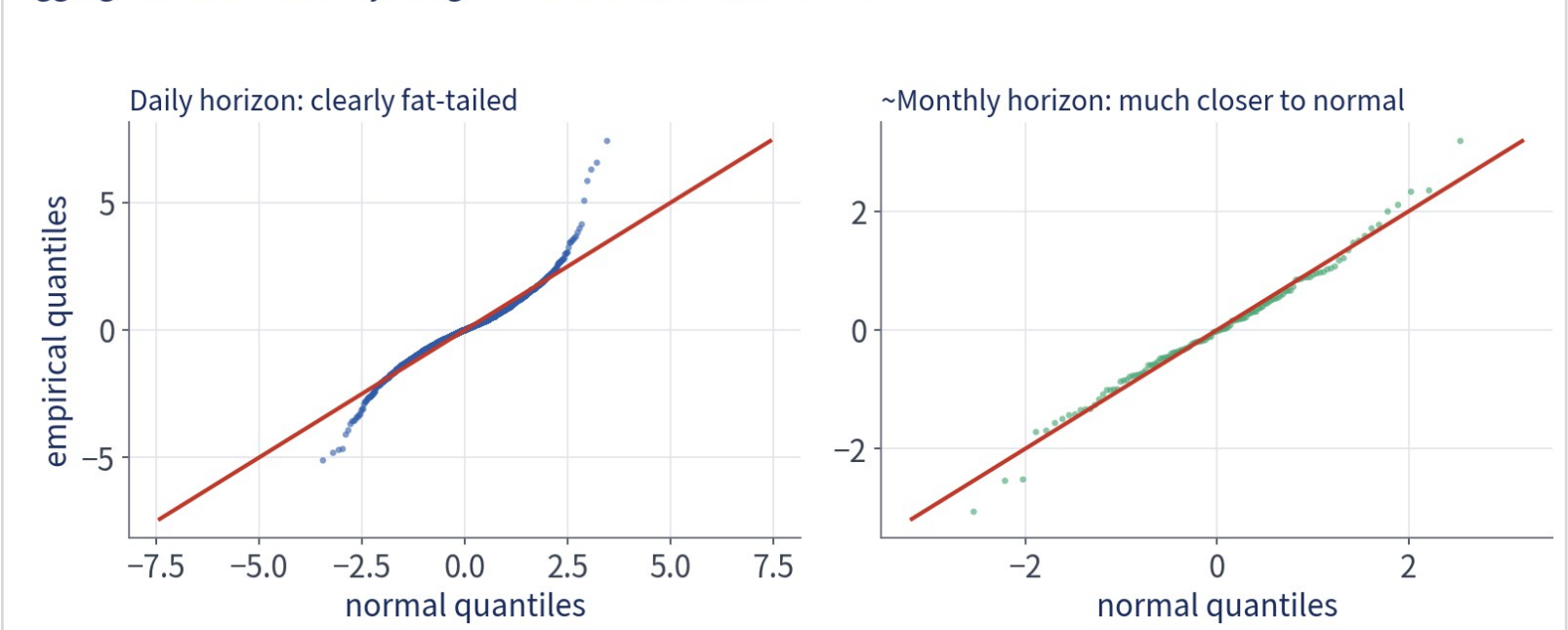
Aggregational Gaussianity

As the horizon lengthens, returns look more and more normal.

What to notice

- Daily: strongly fat-tailed.
- Monthly: nearly normal.
- Summing many returns ...
- ... pulls toward a normal.
- So the horizon matters.

Aggregational Gaussianity: longer horizons look more normal



The Six Stylized Facts Together

Three describe the shape of the distribution; three describe its dynamics through time. Every model is judged by how many it reproduces.

The Six Stylized Facts a Model Must Explain

Blue = dynamics through time | Green = shape of the distribution

1 No autocorrelation in returns

Returns are not linearly predictable.

2 Fat tails (excess kurtosis)

Extremes are far likelier than a normal predicts.

3 Gain / loss asymmetry

Crashes are sharper than rallies (negative skew).

4 Volatility clustering

Big moves cluster; today's vol predicts tomorrow's.

5 Leverage effect

Bad news raises future volatility more than good.

6 Aggregational Gaussianity

Longer-horizon returns look more normal.

From Facts to Models

Each fact motivates a specific tool later in the course

Fat tails

→ heavy-tailed

Replace the normal with a Student-t innovation.

Vol. clustering

→ GARCH

Let today's variance depend on yesterday's shock and variance.

Leverage

→ GJR / EGARCH

Let negative shocks raise variance more than positive ones.

No return memory

→ mean model

Keep the mean simple; put the structure in the variance.

The rest of the course is, in large part, a toolkit for reproducing these six facts.

Three Misconceptions to Avoid

Where intuition quietly goes wrong

Common myths

- "Zero autocorrelation means returns are independent."
- "Volatility clustering lets me predict which way prices go."
- "Fat tails are about a big average return."

The reality

- Uncorrelated $\neq$ independent — the squares are still dependent.
- You can forecast the SIZE of moves, never their SIGN.
- Tails are about the odds of extremes, not the center.

Hunting the Stylized Facts

You describe the goal; Claude designs and runs the exact tests

GOAL Confirm the six stylized facts in your own data using a META-PROMPT — you state what to check, Claude picks the methods.

[INPUT] Meta-prompt · Claude Code

```
Using returns.csv from Lab 1, write and run
stylized_facts.py that checks the facts we
covered this part:
- skewness & kurtosis vs the normal's 0 and 3;
- histogram and Q-Q plot vs a fitted normal;
- ACF of returns vs ACF of squared returns;
- how kurtosis changes at weekly / monthly
  horizons (aggregational Gaussianity).
For each, print the number AND one sentence on
what it implies, then write a short markdown
'stylized-facts report'.
```

Why a meta-prompt

- You give the GOAL; Claude chooses each test.
- Check every result against Part 2's six facts.
- Ask Claude to explain any test you don't know.
- Then refine: 'add a Student-t fit', etc.

Deliverable: stylized_facts.py + a one-page stylized-facts report on your asset.

Three Questions Before You Move On

If you can answer these, Part 2 has landed

Q1

Daily returns show ~zero autocorrelation but squared returns do not. Contradiction?

No — vol clusters

Q2

Your risk model assumes normal returns. What is the main danger?

Understates tails

Q3

Can volatility clustering help you predict tomorrow's return direction?

No — only size

All three follow directly from the six stylized facts.

Three Things to Carry Forward

Part 2 in one screen

1

Returns are not normal, and not i.i.d.

They have fat tails, negative skew, and volatility that clusters. The normal-i.i.d. picture fails on every count.

2

Remember the split: direction vs magnitude

Return direction is unpredictable (efficiency); return magnitude is predictable (volatility has memory). This split organizes the whole course.

3

Facts are the scoreboard for models

Fat tails → Student-t; clustering → GARCH; leverage → GJR/EGARCH. A model earns its keep by reproducing these facts.

Next — Part 3: Random Walks, Martingales, and the Efficient Market Hypothesis.

★ WEEK 1 · PART 2 · PRACTICE

Stylized Facts, Worked by Hand

Measure the Moments, Then Split Direction From Size

Fat tails, negative skew, and volatility clustering

Part 2's empirical regularities, with the arithmetic shown. We compute the skewness and kurtosis of a return sample - finding fat tails and a left skew - then split the autocorrelation of returns from that of squared returns, the split that says direction is unpredictable but volatility is not.

2P

PART 2 · APPLICATIONS

What Returns Actually Look Like

Case A: compute the four moments of a return sample and find fat tails and negative skew - the crash-prone shape real returns take. Case B: contrast the autocorrelation of returns, which is near zero, with that of squared returns, which persists - volatility clustering. Then a Claude Code lab.

What Part 2 Told Us

Now we apply it - with numbers

1

Returns are fat-tailed and left-skewed

Extreme moves happen far more than a normal predicts, and losses cluster in the left tail - markets fall faster than they rise.

2

Direction is unpredictable, size is not

The autocorrelation of returns is near zero, but the autocorrelation of squared returns persists - volatility comes in waves.

3

These are robust regularities, not theorems

Fat tails, negative skew, volatility clustering, and the leverage effect show up across markets and eras.

Everything here is a calculator exercise: one set of moments, one autocorrelation split.

How These Cases Work

Data, then arithmetic, then meaning

Each case has three steps

- Start from a return sample.
- Compute moments, or autocorrelations.
- Then read the regularity.

What you should watch

- How excess kurtosis flags fat tails.
- How negative skew flags crashes.
- How squared returns reveal clustering.

The Shape of Returns

Fat tails and a left skew

A normal distribution has skewness zero and kurtosis three. Real returns don't. We take a return sample with one sharp crash, compute all four moments, and find a strongly negative skewness and a kurtosis well above three - fat tails. These two numbers capture why extreme losses are both more common and more severe than a bell curve suggests.

The sample

- Ten daily returns, in percent.
- Nine small moves, one -5.5% crash.
- Typical of a real return series.

What we'll do by hand

- Compute mean and variance.
- Compute skewness and kurtosis.
- Compare to the normal's 0 and 3.

The Third and Fourth Moments

Asymmetry and tail weight

$$\text{skew} = \frac{\overline{(r-\mu)^3}}{\sigma^3}, \quad \text{kurt} = \frac{\overline{(r-\mu)^4}}{\sigma^4} \quad (\text{normal: skew } 0, \text{ kurt } 3)$$

Skewness

- The average cubed deviation, standardized.
- Negative means a long left tail.
- Crashes outweigh melt-ups.

Kurtosis

- The average fourth-power deviation, standardized.
- Above three means fat tails.
- Extremes are more likely than normal.

The Four Moments

From the sample, by hand

moment	value	normal	reading
mean	-0.50%	-	slight loss
std dev	1.72%	-	the spread
skewness	-2.35	0	left-skewed
kurtosis	7.13	3	fat-tailed

The one crash dominates the higher moments: it drags the mean negative, pulls the skewness to -2.35, and lifts the kurtosis to 7.13 - more than double the normal's three. A single extreme observation reshapes the whole distribution.

Fat Tails and Negative Skew

What the numbers say

$\text{skew} = -2.35$ (< 0 : crashes), $\text{kurt} = 7.13 \Rightarrow \text{excess} = 4.13$ (> 0 : fat tails)

The excess kurtosis

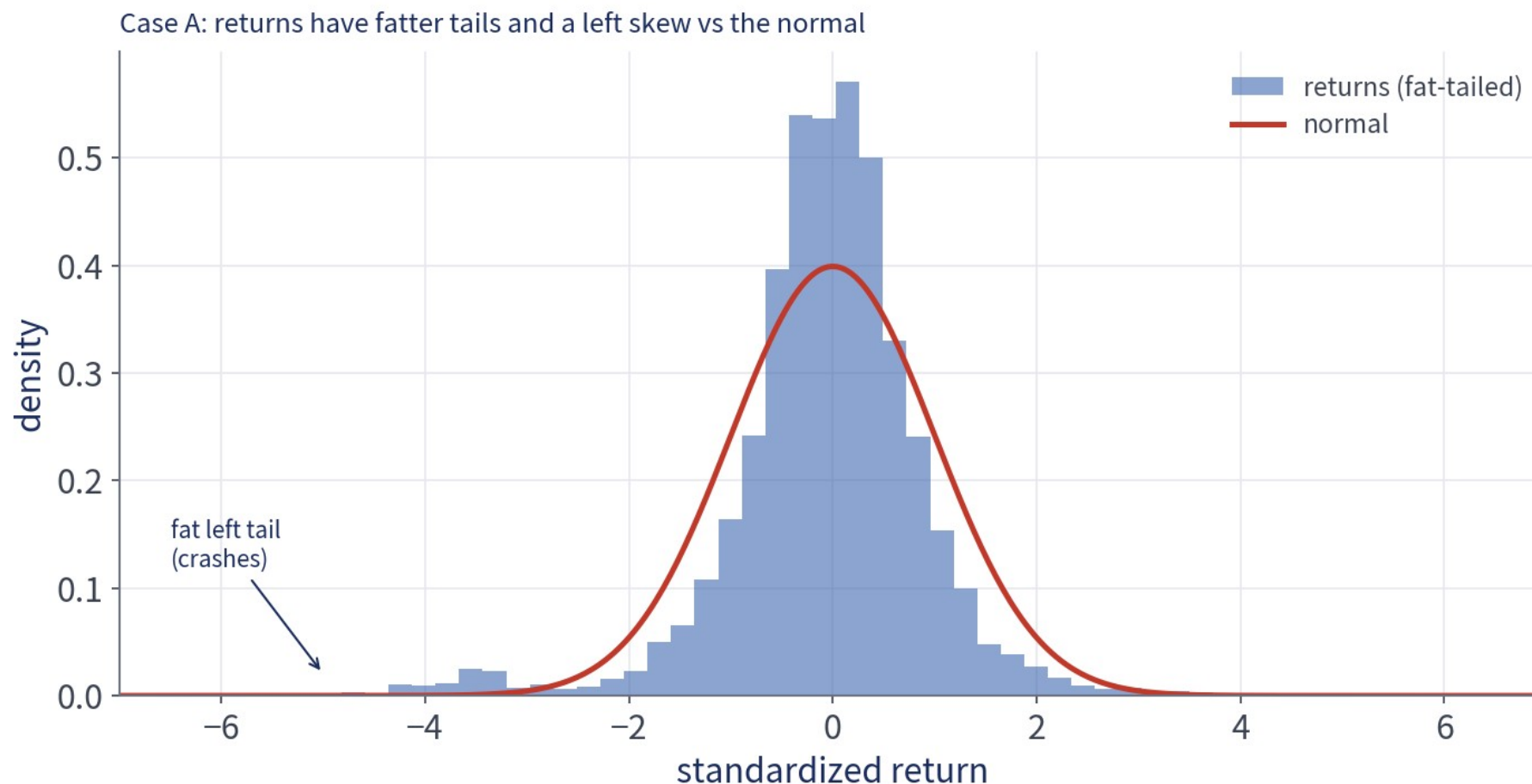
- Kurtosis 7.13 minus the normal's 3.
- An excess of 4.13, well above zero.
- Fat tails: extremes are common.

The negative skew

- A skewness of -2.35.
- The left tail is longer and heavier.
- Markets fall faster than they rise.

Fatter Tails, Longer Left

The return histogram sits taller in the center and, crucially, spills further into the tails than the red normal curve - especially on the left, where crashes live. That extra tail mass is the excess kurtosis; the leftward lean is the negative skew. A normal distribution would badly understate the risk of a large loss.



Why the Shape Matters

Risk lives in the tails

The danger of assuming normal

- A normal model prices tail risk far too cheaply.
- Six-sigma days should never happen - yet they do.
- Value-at-risk built on normality fails.

How heavy the tails are

- The tail follows a power law, not an exponential.
- A tail index around three to four is typical.
- The mean exists; higher moments barely do.

Direction vs Size

The autocorrelation split

Here is the course's most important split. Take a return series that runs calm, bursts into turbulence, then calms again. The autocorrelation of the returns themselves is near zero - you can't predict tomorrow's direction. But the autocorrelation of the squared returns is strongly positive - big moves follow big moves. Direction is unpredictable; size is not.

The series

- Twelve returns: calm, a burst, calm.
- The burst mixes signs.
- But the magnitudes cluster.

What we'll do by hand

- Compute the lag-1 autocorrelation of returns.
- Then of the squared returns.
- See the two diverge.

The Autocorrelation Function

Does the past predict the future?

$$\rho(k) = \frac{\sum_t (r_t - \mu)(r_{t-k} - \mu)}{\sum_t (r_t - \mu)^2}; \quad \text{band} = \pm \frac{1.96}{\sqrt{n}} \text{ (noise if inside)}$$

What it measures

- The correlation of the series with its own past.
- At each lag k , a single number.
- Inside the band means indistinguishable from noise.

The two versions

- On returns: tests direction.
- On squared returns: tests size.
- They tell very different stories.

The Split, Computed

Returns vs squared returns

$$\rho_r(1) = -0.15 \approx 0 \text{ (direction: no memory), } \rho_{r^2}(1) = +0.60 > 0 \text{ (size: clusters)}$$

Returns: no memory

- The lag-1 autocorrelation is about -0.15.
- Inside the noise band - effectively zero.
- Tomorrow's direction is unpredictable.

Squared: strong memory

- The lag-1 autocorrelation is about +0.60.
- Well outside the band.
- Big moves cluster: volatility persists.

The Key Idea

Returns unpredictable, volatility is not

returns unpredictable, volatility is not : $\rho_r(k) \approx 0$ but $\rho_{r^2}(k) > 0$ decaying slowly

Why they differ

- The signs of the burst cancel in returns.
- But squaring makes every big move positive.
- So the clustering survives in the squares.

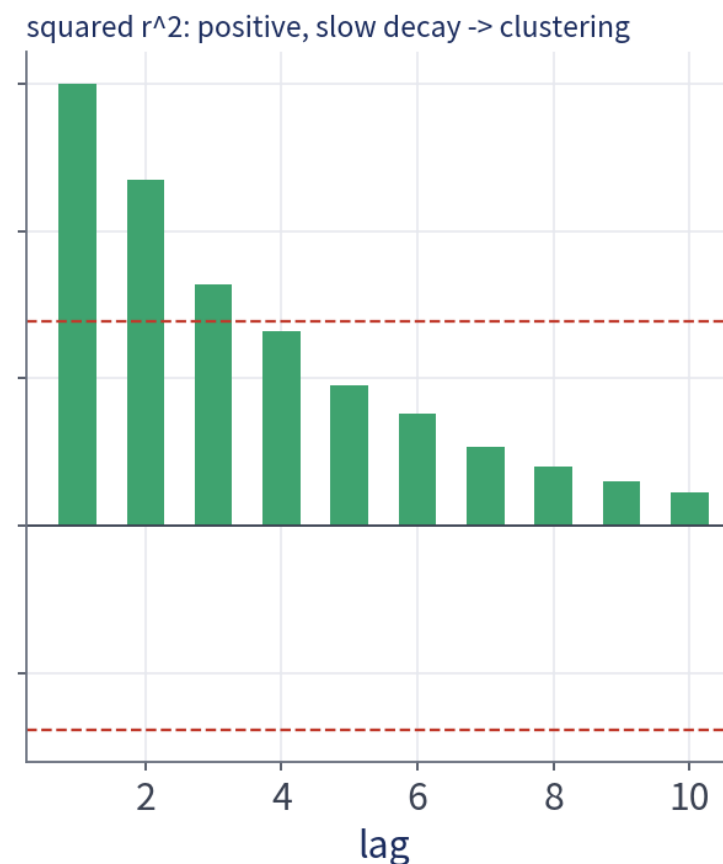
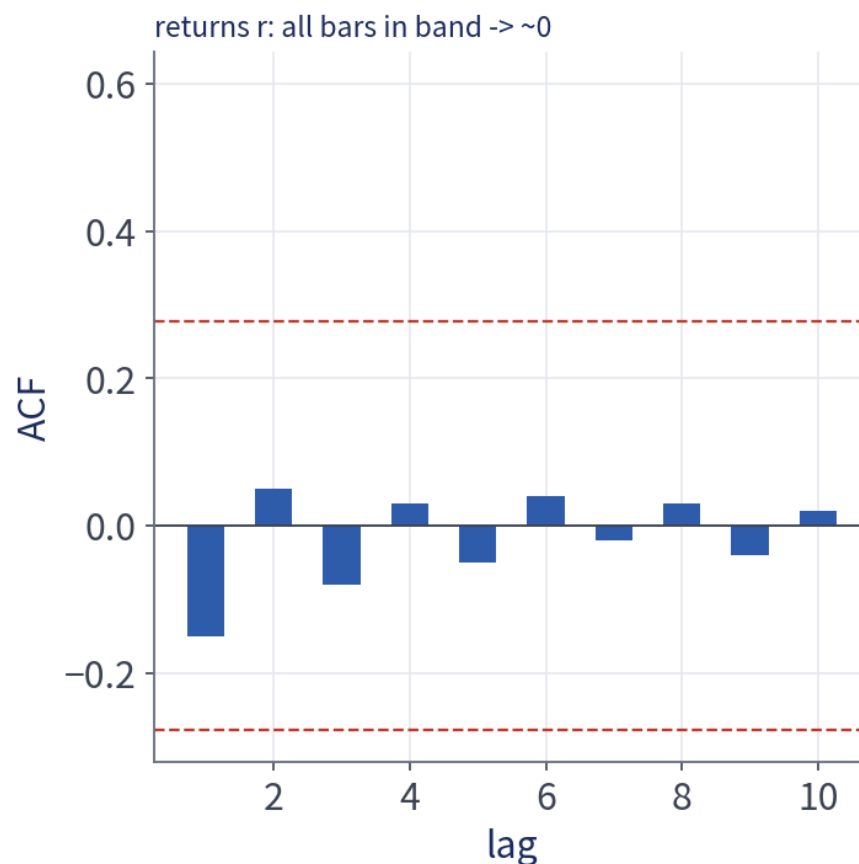
The leverage effect

- Volatility rises more after falls than rises.
- Bad news bites harder than good news.
- A related, direction-linked asymmetry.

Two Autocorrelations, Two Verdicts

Left, the autocorrelation of returns: every bar sits inside the significance band, so direction is unpredictable. Right, the autocorrelation of squared returns: strongly positive and slow to decay, so volatility clusters. This single contrast - flat on the left, persistent on the right - is the engine behind every volatility model to come

Case B: no memory of direction, strong memory of size



What the Split Enables

And what fades with horizon

Why it's the key idea

- You can't forecast direction - markets are efficient.
- But you can forecast volatility - it clusters.
- Every GARCH model exploits exactly this.

Aggregational Gaussianity

- Daily returns are sharply non-normal.
- But monthly or yearly returns look more normal.
- Averaging washes out the fat tails.

The Stylized-Facts Checklist

What every return series shows

These regularities recur across markets and decades. Returns are fat-tailed and negatively skewed, so tail risk dwarfs what a normal predicts. Their autocorrelation is near zero - direction is unpredictable - but the autocorrelation of squared returns persists, so volatility clusters and can be forecast. Bad news raises volatility more than good news, and the non-normality fades as the horizon lengthens. Together these facts set the agenda for the whole course.

The moments (Case A)

- Skewness -2.35, kurtosis 7.13.
- Excess kurtosis 4.13 - fat tails.
- Negative skew - crashes.

The split (Case B)

- Returns autocorrelation about -0.15.
- Squared about +0.60 - clustering.
- Direction unpredictable, size not.

Reproduce the Moments and the Split

Direct Claude Code to check your arithmetic

GOAL Compute the four moments, then the autocorrelation of returns vs squared returns.

[INPUT] Prompt · Claude Code

Write and run `part2_practice.py` that:

CASE A - moments:

- * for returns `[0.5,-0.4,0.6,-0.5,0.3,-0.6,0.4,-0.3,0.5,-5.5]`, print mean, sd, skewness, kurtosis;
- * report the excess kurtosis.

CASE B - the split:

- * for a calm-burst-calm series, print the lag-1 autocorrelation of the returns and of the squares;
- * confirm returns ~ 0 but squares > 0 .

Then plot the return histogram and the two ACFs.

Check against the slides

- skewness -2.35, kurtosis 7.13.
- excess kurtosis 4.13.
- returns lag-1 ~ -0.15 .
- squared lag-1 $\sim +0.60$.

Deliverable: `part2_practice.py` whose printed numbers match the moments and the autocorrelation split.

Three Questions on the Cases

If you can answer these, the practice has landed

Q1 What does excess kurtosis above zero mean?

Fat tails - extremes are more likely than normal

Q2 What does negative skewness indicate?

A longer left tail - crashes outweigh melt-ups

Q3 Why square the returns before the ACF?

Signs cancel; squaring reveals volatility clustering

All three come from the moment calculations and the autocorrelation split.

Three Pitfalls to Avoid

Where stylized facts trip people

Common mistakes

- "Returns are normally distributed."
- "Zero autocorrelation means nothing is predictable."
- "Skewness and kurtosis are minor details."

The reality

- They're fat-tailed and skewed - tails matter most.
- Direction isn't, but volatility is predictable.
- The higher moments drive tail risk.

Two Calculations to Remember

Part 2 applications in one screen

1

The moments reveal fat tails and skew

The return sample had a skewness of -2.35 and a kurtosis of 7.13 - an excess kurtosis of 4.13 over the normal's three. Fat tails and a heavy left tail: extreme losses are both more frequent and more severe than a bell curve would ever predict.

2

Direction is unpredictable, size is not

The lag-1 autocorrelation of returns was about -0.15, inside the noise band, so tomorrow's direction can't be forecast. But for squared returns it was about +0.60 - big moves follow big moves. Volatility clusters, and that is forecastable.

3

These facts set the course's agenda

Fat tails demand non-normal risk models; volatility clustering motivates GARCH; the leverage effect and aggregational Gaussianity refine both. Every later week answers a stylized fact seen here.

Next — Part 3 practice: market efficiency and the random walk, why direction resists prediction.

★ WEEK 1 · PART 3 of 3

Random Walks & Market Efficiency

Is the Past Any Guide to the Future?

Random walk · martingale · the Efficient Market Hypothesis · anomalies

Part 1 gave us returns; Part 2 gave us their patterns. Part 3 asks the deepest question: how much of the future can we actually predict — and why the market resists us.

3

PART 3 · EFFICIENCY

Why Markets Resist Prediction

From the random walk to the Efficient Market Hypothesis — the theory of why beating the market is so hard, and the evidence for and against it.

The Mathematician Who Beat the Market

Edward O. Thorp — a story that frames the whole debate

A math professor first beat casino blackjack with card counting, then turned the same rigor on Wall Street — and quietly outperformed for nearly two decades.

The setup

- Thorp proved blackjack was beatable — casinos changed their rules.
- He then asked: are financial markets beatable too?
- His tool: quantify mispricing in warrants and options, and hedge it.

The claim to test

- The reigning academic view: prices follow a random walk.
- If true, no strategy should beat the market on a risk-adjusted basis.
- So who is right — the theory, or the professor?

The Resolution, and the Lesson

What Thorp's record actually teaches us

His fund Princeton/Newport Partners ran 1969–1988, using hedged warrant and option arbitrage to earn strong, remarkably steady returns with no losing year.

So markets can be beaten?

- Yes — but only through narrow, identifiable, hedged edges.
- Not by predicting direction; by pricing risk better than others.
- And such edges shrink as more people find them.

The lesson for us

- Efficiency is a powerful baseline, not an absolute law.
- The bar for 'an edge' is very high — most apparent edges are noise.
- This tension runs through the entire course.

The Central Question of Part 3

Where, exactly, does prediction become possible?

1

If prices already reflect all information, the past is useless

That is the core intuition of market efficiency: whatever is knowable is already in the price, so the next move is essentially unpredictable.

2

The random walk and the martingale make this precise

These two models formalize 'unpredictable.' We will see exactly what each one assumes — and how they differ.

3

Reality is messier than the theory

Anomalies, bubbles, and Thorp's own record show the theory is an approximation. The debate is about how good an approximation.

Efficiency is the null hypothesis; the whole field is a series of attempts to reject it.

The Random Walk Hypothesis

Tomorrow's price = today's price + drift + an unpredictable shock

$$P_t = \mu + P_{t-1} + \varepsilon_t, \quad \varepsilon_t \text{ i.i.d.}, \quad \mathbb{E}[\varepsilon_t] = 0$$

What it says

- The best forecast of tomorrow's price is today's price (plus drift).
- Shocks are unpredictable with zero mean — no exploitable pattern.
- In returns: $r_t = \text{drift} + \text{noise}$, with no memory.

Why it matters

- It is the simplest model of an unpredictable price.
- It implies technical analysis of past prices cannot add value.
- It is the benchmark every 'edge' must beat.

What a Random Walk Looks Like

Nine simulated paths from an identical starting point. Each looks like it has 'trends' — yet none was foreseeable. Our eyes invent patterns that were never there.

Nine random walks from the same start — each equally plausible, none foreseeable



Three Flavors of Random Walk

'Unpredictable' comes in strengths — and only the weakest fits real data

Version	Shocks (increments) are ...	What it rules out
RW1 (strongest)	independent AND identically distributed	any dependence whatsoever
RW2	independent, but not identically distributed	any dependence; allows a changing mean/variance
RW3 (weakest)	merely uncorrelated (may still be dependent)	only LINEAR dependence

Key link to Part 2: real returns fit only RW3 — uncorrelated, yet dependent through volatility clustering.

The Martingale Property

A precise statement of 'a fair game'

$$\mathbb{E}[P_{t+1} \mid \mathcal{F}_t] = P_t \quad (\text{martingale: a fair game})$$

Reading the notation

- $\mathcal{F}_t$ is the information available today.
- Given everything known now, the expected future price is today's.
- So no strategy using today's info earns an expected edge.

Fair game intuition

- A martingale is the math of a bet with no expected gain.
- Excess returns form a 'martingale difference': mean zero given the past.
- This is the cleanest formal version of market efficiency.

Martingale vs Random Walk

A subtle but important difference

Random walk (stronger)

- Restricts the WHOLE distribution of shocks (independence).
- Rules out any dependence, including in the variance.
- Too strong: real returns violate it via volatility clustering.

Martingale (weaker)

- Restricts only the CONDITIONAL MEAN (expected next move = 0).
- Says nothing about variance — clustering is allowed.
- This is why efficiency is usually stated as a martingale, not a random walk.

From Random Walk to Market Efficiency

The economic idea behind the mathematics

Prices are unpredictable not by accident, but because competition among informed traders forces all knowable information into the price almost instantly.

The mechanism

- Any predictable profit attracts traders who trade it away.
- Their trading moves the price until the profit disappears.
- What remains is an unpredictable, 'fair-game' residual.

The consequence

- Prices reflect information — that is the Efficient Market Hypothesis.
- The random walk is the statistical shadow of this economic force.
- Now we make 'reflect information' precise.

The Efficient Market Hypothesis (EMH)

Prices fully reflect the available information set

$$\mathbb{E}[r_{t+1} - \mathbb{E}(r_{t+1} | I_t) | I_t] = 0 \quad (\text{prices reflect information})$$

In words

- You cannot systematically beat the market using information in I_t .
- Whatever is in I_t is already priced; only surprises move prices.
- By definition, surprises are unpredictable.

The key subtlety

- Efficiency is always relative to an information set I_t .
- Different information sets give different, testable forms (next).
- 'Efficient' never means 'the price is correct' — only 'unbeatable'.

Three Forms of Market Efficiency

Each form widens the information set already priced in. The wider the set, the harder the market is to beat.

Three Forms of Market Efficiency

Each form widens the information set already priced in. Bigger set = harder to beat the market.



What each form says you CANNOT do:

- **Weak:** beat the market using charts of past prices.
- **Semi-strong:** profit from public news — it is already in the price by the time you trade.
- **Strong:** profit even on inside information (rejected in practice — insiders do earn abnormal returns).

Most evidence supports weak & semi-strong; strong form is genera

Testing Weak-Form Efficiency

Can past prices predict future returns? Three classic tests

$$VR(q) = \frac{\text{Var}(r_t(q))}{q \cdot \text{Var}(r_t)} = 1 \quad \text{under a random walk}$$

The tests

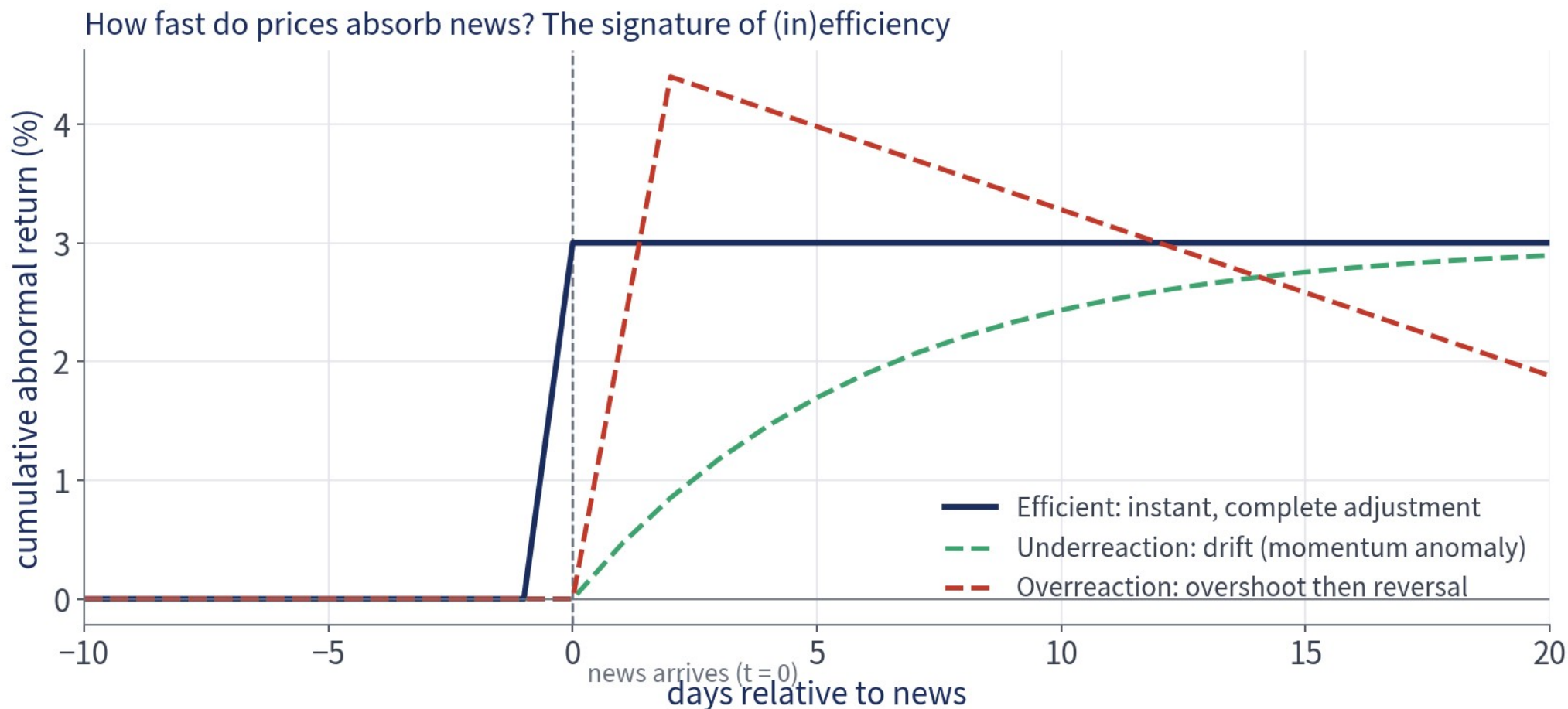
- Autocorrelation: is $\rho_R(k)$ different from zero? (Part 2: no.)
- Variance ratio: does variance scale linearly with horizon?
- Runs test: are up/down streaks more ordered than chance?

The verdict

- Under a random walk, the variance ratio equals 1 at all horizons.
- Broadly, weak-form efficiency holds well for liquid markets.
- Deviations exist but are small and often unstable.

Testing Semi-Strong Form: How Fast Do Prices React?

An efficient market jumps once, instantly and completely (navy). Underreaction (drift) and overreaction (reversal) are the footprints of inefficiency.



The Joint Hypothesis Problem

Why efficiency can never be tested on its own

$$\underbrace{\text{test of efficiency}}_{\text{what we want}} = \underbrace{\text{efficiency}}_{\text{joint hypothesis}} \text{ and } \underbrace{\text{a model of } \mathbb{E}[r]}_{\text{joint hypothesis}}$$

Fama's insight

- To test efficiency you need a model of 'normal' expected returns.
- So every test is really a test of TWO things at once.
- A rejection could mean inefficiency — or a wrong return model.

Why it matters

- 'Anomalies' may just be a mispriced risk factor we haven't modeled.
- This makes the efficiency debate genuinely hard to settle.
- It keeps both camps honest — and the question open.

Evidence That Markets Are Efficient

Why the baseline is so hard to beat

What we observe

- Most active managers underperform simple index funds after fees.
- Persistent, repeatable alpha is rare — winners rarely stay winners.
- Prices react to public news within seconds, not days.

Why it holds

- Millions of competing, informed traders leave little on the table.
- Low-cost information and fast execution close gaps quickly.
- The bar Thorp had to clear shows just how efficient markets are.

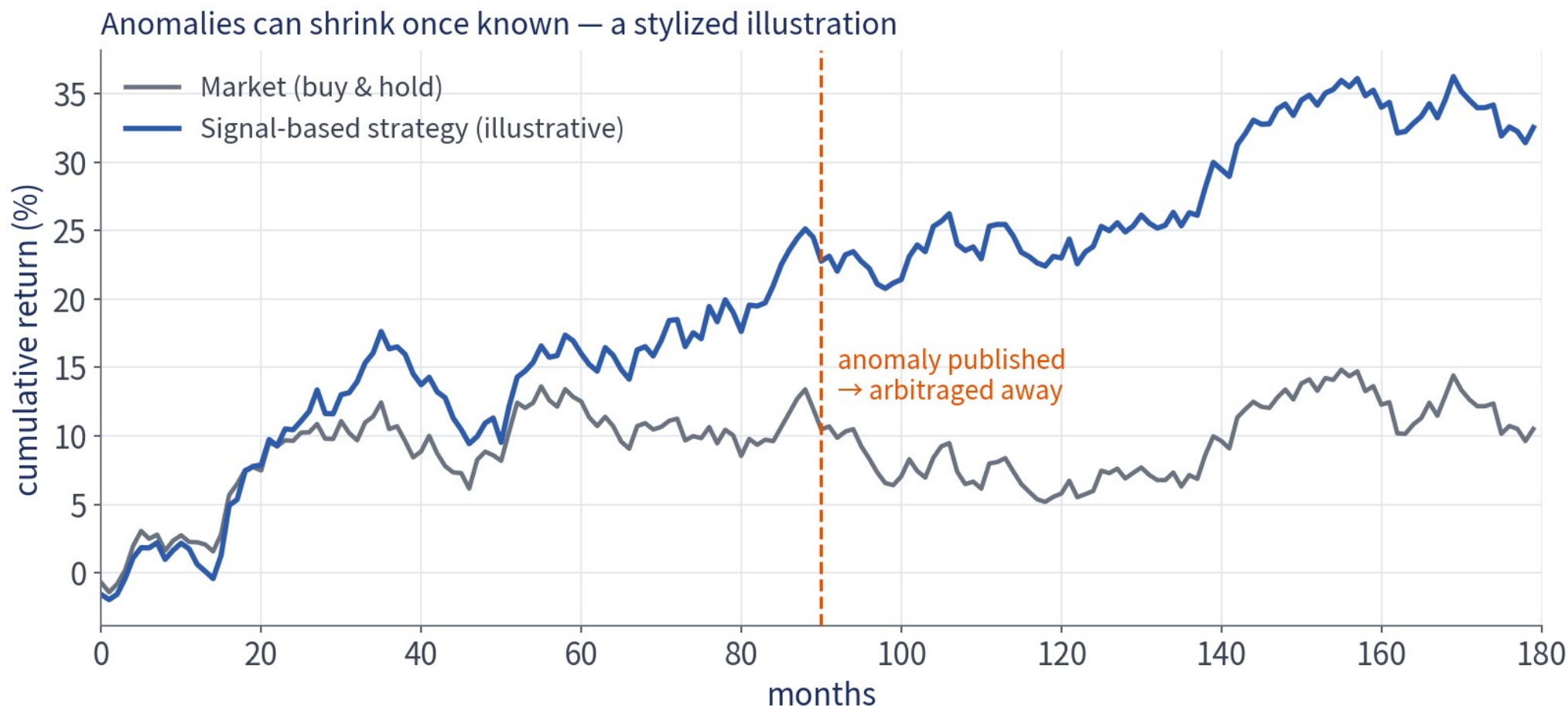
The Anomalies

Patterns the strict theory struggles to explain

Anomaly	The observed pattern	Efficient-market reply
Momentum	Past winners keep winning (3–12 months)	A risk premium — or it fades once known
Value	Cheap (low P/B) stocks tend to outperform	Compensation for distress risk
Size	Small-cap stocks historically outperform	Weak and unstable after publication
Calendar effects	Returns cluster in certain periods	Often vanish out of sample
Post-earnings drift	Prices drift after earnings surprises	Underreaction plus limits to arbitrage

Anomalies Often Fade Once They Are Known

A stylized illustration: a profitable signal loses its edge after publication, as arbitrageurs pile in — exactly what an efficient market predicts.



The Behavioral Challenge

Where human psychology bends prices

Systematic biases

- Overreaction: crowds chase, then reverse (bubbles and busts).
- Underreaction: news is absorbed slowly, creating drift.
- Herding, anchoring, and loss aversion push prices off value.

Why mispricing can persist

- Limits to arbitrage: correcting a mispricing is costly and risky.
- 'Markets can stay irrational longer than you can stay solvent.'
- So biases are not always competed away immediately.

The Grossman–Stiglitz Paradox

A market cannot be perfectly efficient

If prices already reflected all information, no one would be paid to gather it — but then who puts the information into prices? Perfect efficiency is self-defeating.

The logic

- Gathering information is costly.
- In a perfectly efficient market, that effort earns nothing.
- So rational traders would stop gathering — and efficiency collapses.

The resolution

- Markets are 'efficiently inefficient': just enough profit to pay analysts.
- A small, hard-won edge is the reward for making prices informative.
- This is precisely the space Thorp operated in.

The Adaptive Markets Hypothesis

Efficiency as an evolving, not fixed, property

The idea (Lo)

- Efficiency is not binary; it waxes and wanes with conditions.
- Investors adapt, strategies compete, and edges are born and die.
- It blends efficient-market logic with behavioral realism.

What it implies

- Anomalies can appear, be exploited, and disappear over time.
- No strategy works forever; adaptation is the only constant.
- This reconciles Thorp's edge with the market's overall efficiency.

Efficiency vs Reality

The debate that never fully closes

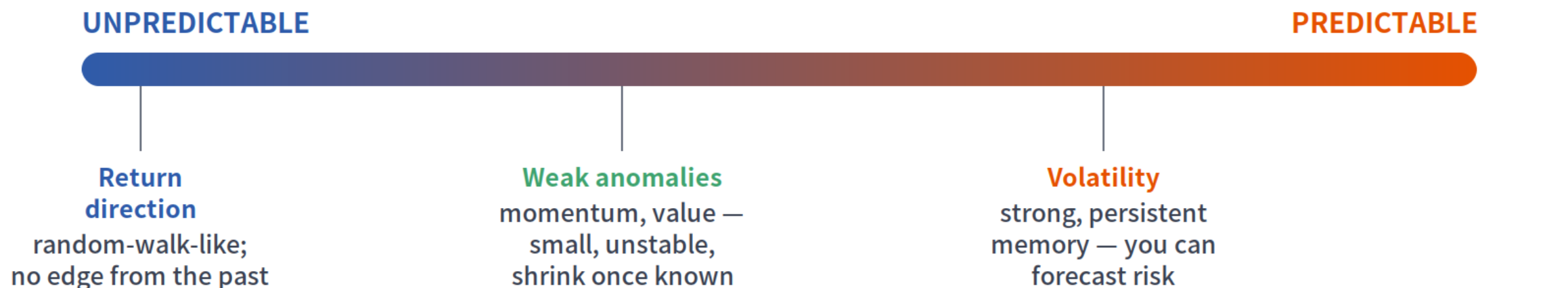
Question	Efficient-market view	Behavioral / reality view
Can you beat the market?	No — prices are fair, net of costs	Sometimes, via narrow identifiable edges
Why do anomalies exist?	Risk premia, or data mining	Real mispricing from human bias
Are investors rational?	On average and in aggregate, yes	Systematically biased at times
Do prices always reflect value?	Yes, and quickly	Bubbles and crashes suggest not

The Predictability Spectrum

Efficiency is not all-or-nothing. Return direction sits near the unpredictable end; volatility sits near the predictable end — the throughline of the whole week.

The Predictability Spectrum

Efficiency is not all-or-nothing. Different quantities sit at different points on this line.



Takeaway: the DIRECTION of returns is near the unpredictable end; the SIZE of returns (volatility) is near the predictable end.

Putting Efficiency to the Test

Direct Claude Code to run the weak-form efficiency tests on real data

GOAL Test weak-form efficiency on your own returns — and see the joint-hypothesis caveat in action.

[INPUT] Prompt · Claude Code

Using returns.csv, write and run efficiency.py:

1. autocorrelation of returns at lags 1..20 with 95% confidence bands;
 2. the variance ratio $VR(q)$ for $q = 2, 5, 10, 20$, testing whether $VR(q) = 1$;
 3. a runs test on the sign of daily returns.
- For each test, print the statistic and a plain-English verdict on weak-form efficiency, and state the joint-hypothesis caveat. Save the ACF and variance-ratio plots to /figs.

How to read it

- $VR \approx 1$ and $\rho \approx 0$ support weak-form efficiency.
- A rejection may mean a wrong return model.
- Try an illiquid ticker — do results change?
- Evidence, never proof — stay skeptical.

Deliverable: efficiency.py + a short note on whether your asset looks weak-form efficient.

Three Questions Before You Move On

If you can answer these, Part 3 has landed

Q1

Real returns are uncorrelated yet dependent. Which random-walk version fits?

RW3 (weakest)

Q2

Why can market efficiency never be tested by itself?

Joint hypothesis

Q3

Efficiency says the past is useless. Does that contradict volatility forecasting?

No — size $\neq$ sign

All three tie Part 3 back to the stylized facts of Part 2.

Week 1 in Three Ideas

Pulling the whole week together

1

Speak in returns — and prefer log returns

Part 1: prices are non-stationary; returns are the tractable unit. Match the mean to the question, and respect the traps.

2

Real returns have a fingerprint

Part 2: fat tails, negative skew, and clustering volatility. Direction is unpredictable; magnitude is not.

3

Markets are efficient — but not perfectly

Part 3: the random walk and EMH explain why prediction is hard, while anomalies and Thorp show the theory is an approximation.

Next — Week 2: Stationarity and Unit Roots, making 'roughly stationary' precise.

★ WEEK 1 · PART 3 · PRACTICE

Market Efficiency, Worked by Hand

Test the Random Walk With a Variance Ratio and a Runs Count

Weak-form efficiency, and the limits of testing it

Part 3's central question, with the arithmetic shown. We run two classic weak-form tests - the variance ratio and the runs test - to ask whether past prices predict future returns, then confront why efficiency can never quite be proven: the joint hypothesis problem and the Grossman-Stiglitz paradox.

3P

PART 3 · APPLICATIONS

Can the Past Predict the Future?

Case A: the variance ratio test - under a random walk, variance scales linearly with the horizon, so a ratio above one signals trending. Case B: the runs test counts up-and-down streaks against what randomness predicts - and the deep reasons efficiency resists proof. Then a Claude Code lab.

What Part 3 Told Us

Now we apply it - with numbers

1

The random walk makes a testable claim

Tomorrow's price is today's plus drift plus an unpredictable shock. So return variance should scale linearly with the horizon.

2

Weak-form efficiency has classic tests

Autocorrelation, the variance ratio, and the runs test all ask the same thing: can past prices predict future returns?

3

Efficiency can never be tested alone

The joint hypothesis problem and the Grossman-Stiglitz paradox mean perfect efficiency is untestable - and impossible.

Everything here is a calculator exercise: one variance ratio, one runs count.

How These Cases Work

Data, then arithmetic, then meaning

Each case has three steps

- Start from returns or their signs.
- Compute the ratio, or count the runs.
- Then judge efficiency.

What you should watch

- How variance should scale under a random walk.
- How the runs count compares to its expectation.
- Why a clean verdict is so hard to reach.

The Variance Ratio Test

Does the market follow a random walk?

Under a random walk, returns are independent, so the variance of a k-period return is exactly k times the one-period variance. The variance ratio divides the observed multi-period variance by that prediction. A ratio of one means a random walk; above one signals trending, below one mean reversion. We compute it for a five-day

The inputs

- One-day return variance of 4.
- Observed five-day variance of 26.
- The random-walk prediction is 20.

What we'll do by hand

- Form the ratio of observed to predicted.
- Read whether it exceeds one.
- Interpret the market structure.

The Random Walk's Prediction

Variance scales linearly

$$P_t = P_{t-1} + \mu + \varepsilon_t \Rightarrow \text{Var}(r_k) = k \cdot \text{Var}(r_1) \quad (\text{variance scales linearly with the horizon})$$

Why linear scaling

- Independent shocks add their variances.
- So k periods give k times the variance.
- That's the null the test checks.

The ratio

- Observed variance over k times the one-period.
- Exactly one under a random walk.
- Deviations reveal autocorrelation.

Computing the Ratio

Observed over predicted

$$VR(5) = \frac{\text{Var}(r_5)}{5 \cdot \text{Var}(r_1)} = \frac{26}{5 \times 4} = \frac{26}{20} = 1.30$$

The calculation

- The random walk predicts five times four, or 20.
- The observed five-day variance is 26.
- The ratio is 26 over 20, or 1.30.

First read

- The ratio exceeds one.
- Variance grew faster than linearly.
- That points to positive autocorrelation.

Reading the Ratio

Trending, random, or reverting

$VR = 1$ random walk; $VR > 1$ trending; $VR < 1$ mean-reverting

What 1.30 means

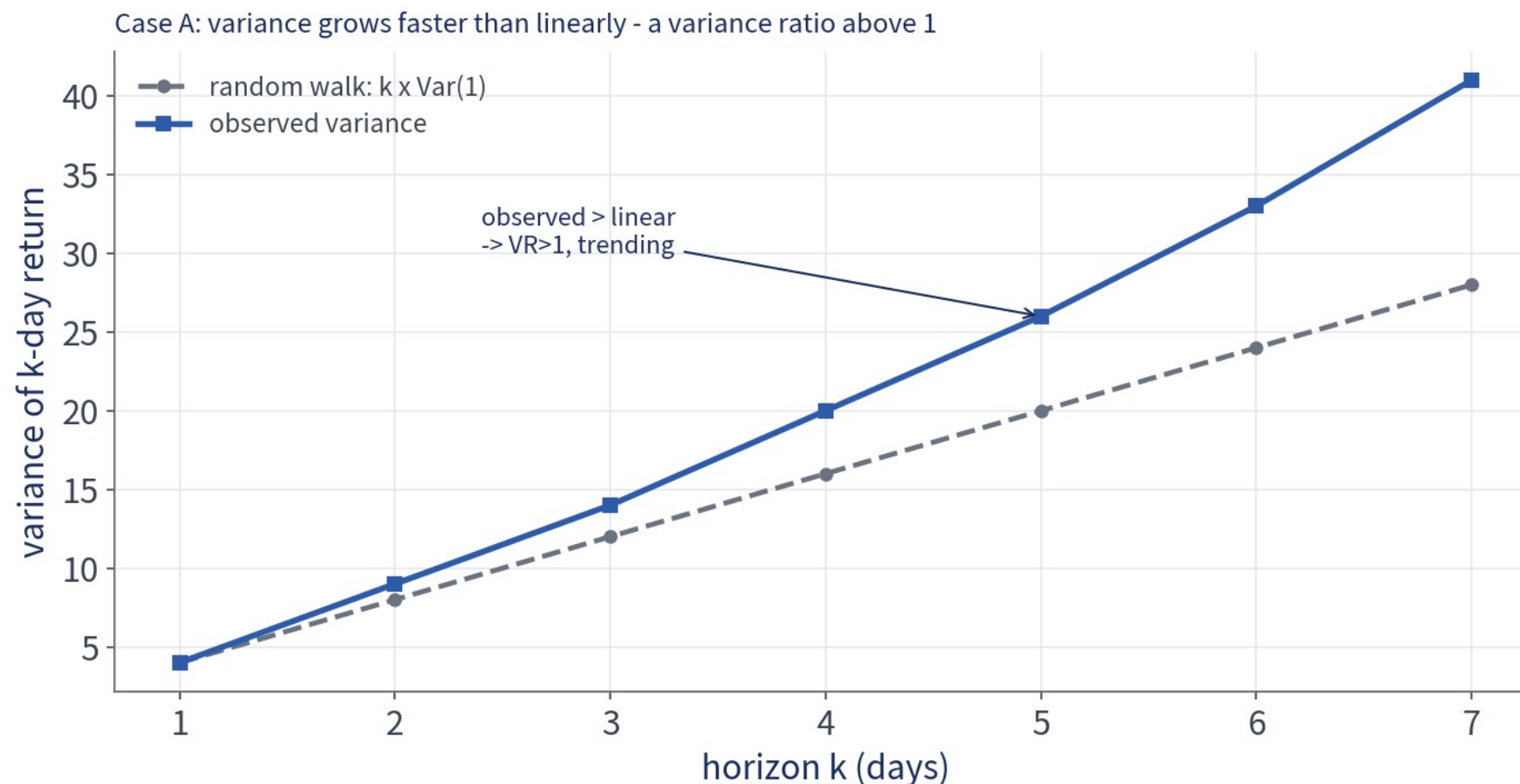
- Above one: returns trend.
- Gains tend to follow gains.
- A mild momentum, a mild inefficiency.

The other cases

- Exactly one: a pure random walk.
- Below one: mean reversion.
- The ratio maps to net autocorrelation.

Faster Than Linear

The dashed line is the random walk's prediction: variance rising in exact proportion to the horizon. The observed variance climbs above it, curving upward - the signature of positive autocorrelation. That gap is the variance ratio exceeding one, and it says the series trends rather than wandering purely at random.



Three Forms of Efficiency

How much is already priced in

The three forms

- Weak: past prices are priced in.
- Semi-strong: all public information is.
- Strong: even private information is.

What the test probes

- The variance ratio tests the weak form.
- A ratio near one supports it.
- Our 1.30 hints at a small, exploitable trend.

The Runs Test and Its Limits

Counting streaks and what proof requires

The runs test asks a simpler question: do up and down days alternate the way pure chance would? We count the runs - streaks of same-sign returns - and compare to the number randomness predicts, via a z-score. Then we face the deeper problem: even a clean result can't prove efficiency, because every test assumes a

The sequence

- Ten days of return signs.
- Five up, five down.
- Grouped into streaks.

What we'll do by hand

- Count the runs.
- Compare to the expected number.
- Compute the z-score and judge.

The Runs Test

Streaks against chance

$$\mathbb{E}[R] = \frac{2n_+n_-}{n} + 1 = 6, \quad z = \frac{R - \mathbb{E}[R]}{\text{sd}} = \frac{5 - 6}{1.49} = -0.67$$

The expectation

- Under randomness, the expected runs is a formula.
- From the counts of ups and downs.
- Here it comes to six.

The test

- Too few runs means trending.
- Too many means mean reversion.
- A z-score measures the gap.

Counting the Runs

Observed against expected

quantity	value
signs	+ + + - - + - - - +
runs observed	5
expected runs	6.0
z-score	-0.67 (not significant)

The ten days form five runs - three ups, two downs, one up, three downs, one up. Randomness predicts six, with a standard deviation of about 1.49, so the z-score is minus 0.67. Well inside plus-or-minus 1.96: we cannot reject weak-form efficiency.

Why Proof Is Elusive

The joint hypothesis and the paradox

$|z| = 0.67 < 1.96 \Rightarrow$ cannot reject efficiency; but a test needs a return model (joint hypothesis)

The joint hypothesis problem

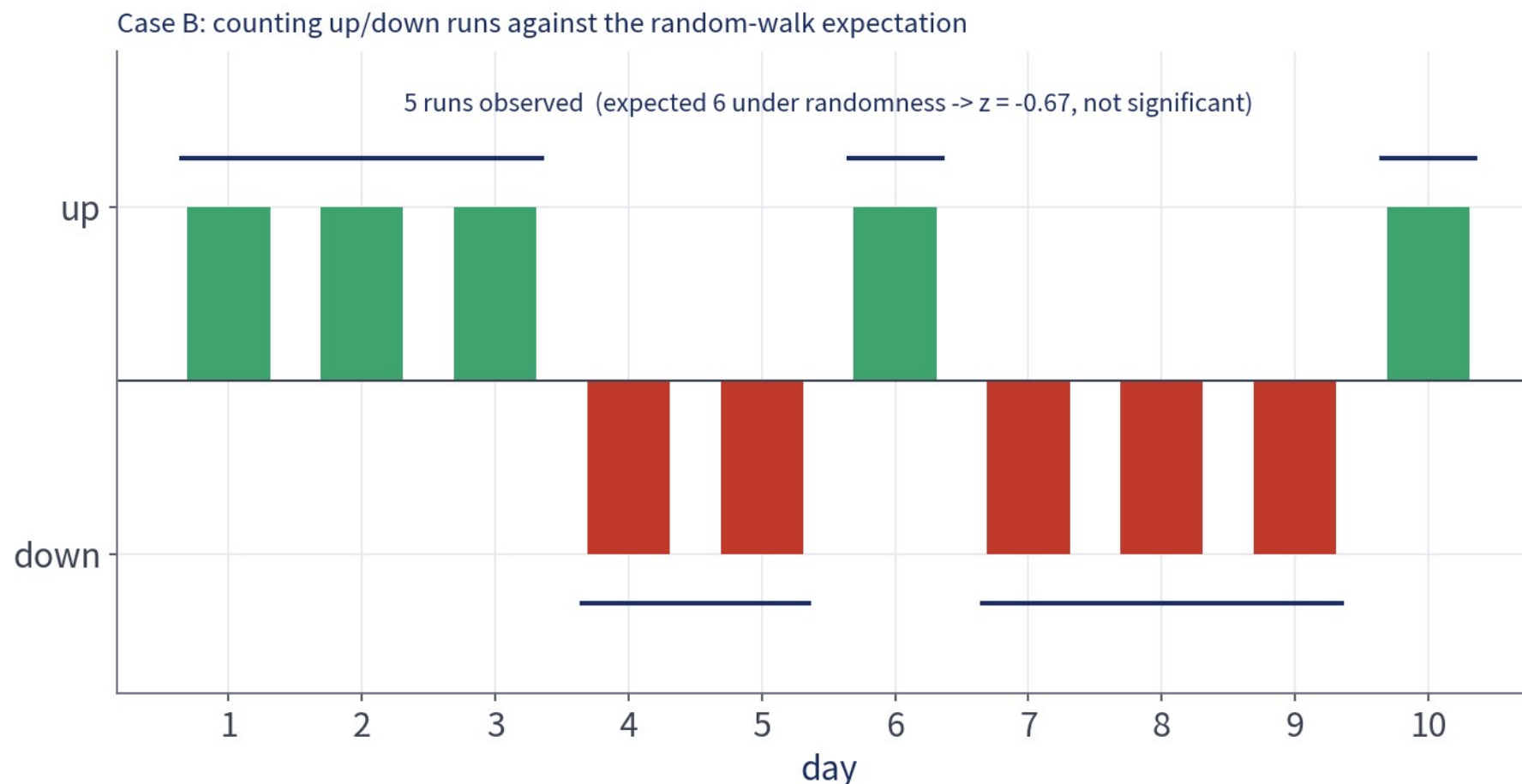
- Any test needs a model of expected returns.
- A rejection could blame either one.
- So efficiency is never tested alone.

Grossman-Stiglitz

- If prices reflected everything, no one would research.
- But then nothing would get priced in.
- Perfect efficiency is self-defeating.

Five Streaks, Six Expected

Up days in green, down days in red, with the runs bracketed above. Five streaks appear where pure chance would produce about six - a difference of one, small against a standard deviation of 1.49. The z-score of minus 0.67 sits comfortably inside the band: this sequence is entirely consistent with a random walk.



Efficiency as a Moving Target

The adaptive view

Why we rarely reject

- Markets are close to weak-form efficient.
- Direction is hard to predict - as Part 2 showed.
- Small edges are quickly arbitrated away.

The adaptive markets view

- Efficiency evolves as participants learn.
- Opportunities appear, get competed away, recur.
- Not a fixed law, but an ecology.

The Efficiency Toolkit

The weak-form tests share one logic: compare what you observe to what a random walk predicts. The variance ratio checks whether variance scales linearly; the runs test checks whether streaks match chance; the autocorrelation test, from Part 2, checks direct predictability. Deviations hint at trending or reversion. But interpret with humility: the joint hypothesis problem and the

The variance ratio (Case A)

- Observed 26 over predicted 20.
- A ratio of 1.30 - trending.
- A small weak-form deviation.

The runs test (Case B)

- Five runs against an expected six.
- z of -0.67 - not significant.
- Consistent with a random walk.

Reproduce the Variance Ratio and Runs Test

Direct Claude Code to check your arithmetic

GOAL Compute a variance ratio, then a runs-test z-score - the two weak-form tests.

[INPUT] Prompt · Claude Code

```
Write and run part3_practice.py that:  
CASE A - variance ratio:  
  * with a 1-day variance of 4 and an observed 5-day  
    variance of 26, print VR(5) and interpret it;  
  * confirm VR = 1 would mean a random walk.  
CASE B - runs test:  
  * for the signs + + + - - + - - +, count the runs;  
  * print the expected runs, the sd, and the z-score;  
  * report whether efficiency can be rejected.  
Then plot variance vs horizon and the sign runs.
```

Check against the slides

- $VR(5) = 26/20 = 1.30$.
- trending ($VR > 1$).
- runs 5, expected 6.
- $z = -0.67$ (not significant).

Deliverable: part3_practice.py whose printed numbers match the variance ratio and the runs test.

Three Questions on the Cases

If you can answer these, the practice has landed

Q1 What does a variance ratio above one mean?

Positive autocorrelation - the series trends

Q2 What does the runs test compare?

Observed streaks against what randomness predicts

Q3 Why can't efficiency be tested alone?

Every test also assumes a model of expected returns

All three come from the variance ratio, the runs test, and the joint hypothesis problem.

Three Pitfalls to Avoid

Where efficiency testing trips people

Common mistakes

- "A variance ratio above one guarantees profit."
- "Failing to reject proves the market efficient."
- "A rejection proves inefficiency."

The reality

- Edges are small and often vanish after costs.
- Failing to reject is weak evidence, not proof.
- It might be the return model that's wrong.

Two Calculations to Remember

Part 3 applications in one screen

1

The variance ratio spots trending

With a one-day variance of 4, a random walk predicts a five-day variance of 20 - but the observed 26 gave a variance ratio of 1.30. Above one means variance grows faster than linearly: positive autocorrelation, a mild trend, a small weak-form deviation.

2

The runs test found nothing to reject

The ten-day sign sequence formed five runs where randomness predicts six, a z-score of minus 0.67 - well inside the band. We cannot reject weak-form efficiency, the usual outcome: direction is genuinely hard to predict.

3

Efficiency is a benchmark, not a proof

The joint hypothesis problem ties every test to a return model, and Grossman-Stiglitz shows perfect efficiency is self-defeating. Treat efficiency as a disciplined baseline - and the adaptive-markets view as how it evolves.

Week 1 complete — returns, their stylized facts, and the efficiency that makes direction so hard to predict.